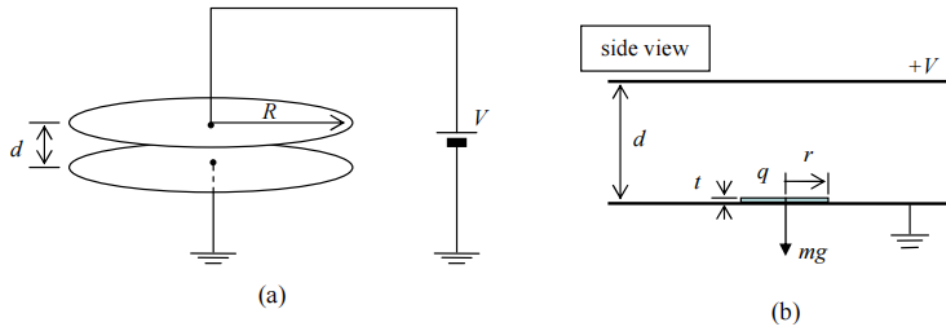


1-nazariy savol: “Ping-Pong” rezistor (ping-pong rezistori)



1.1-rasm: (a) o‘zgarmas kuchlanish manbaiga ulangan parallel plastinkali kondensator (sig‘im) va (b) kondensator ichiga kichik disk qo‘yilgan parallel plastinkalarning yon ko‘rinishi (matn ichida tafsilotlar).

Kondensator (sig‘im saqlovchi ikki plastinkali asbob) radiusi R bolgan ikkita aylana parallel plastinkadan iborat, ular bir-biridan d masofa bilan ajratilgan ($d \ll R$), 1.1(a)-rasmدا korsa-tilganidek. Yuqori plastinka o‘zgarmas kuchlanish manbaiga (V potentsialga) ulangan, pastki plastinka esa yerga ulangan. Keyin massasi m , radiusi r ($\ll R, d$) va qalinligi t ($\ll r$) bo‘lgan yupqa va kichik disk pastki plastinkaning markaziga qo‘yiladi, 1.1(b)-rasmدا ko‘rsatilganidek.

Plastinkalar orasidagi bo‘shliq vakuum (havosiz muhit) deb faraz qilaylik, dielektrik o‘tkazuvchanlik (elektr maydonini o‘tkazish qobiliyati) ϵ_0 ; plastinkalar va disk mukammal o‘tkazgichlardan (zaryadni to‘liq o‘tkazadigan material) yasalgan; barcha elektrostatik chekka effektlar (qir-ralardagi maydon notekisliklari) e‘tiborga olinmasin. Butun zanjirning induktivligi (o‘zinduksiya) va relyativistik effektlarni (nisbiylik nazariyasiga oid ta’sirlar) xavfsiz ravishda e‘tiborsiz qoldirish mumkin. Tasvir zaryad effekti (zaryadning akslanishi) ham e‘tiborga olinmasin.

- [1.2 ball] Diskni plastinkalar orasiga kiritishdan oldin, 1.1(a)-rasmدا ko‘rsatilganidek, d masofa bilan ajratilgan plastinkalar orasidagi elektrostatik kuchni (zaryadlar orasidagi tortishish/itarishish kuchi) F_p hisoblang.
- [0.8 ball] Disk pastki plastinkaga qo‘yilganda, 1.1(b)-rasmда diskдаgi q zaryad kuchlanish V bilan $q = \chi V$ munosabati orqali bog‘langan. χ ni r , d va ϵ_0 larda ifodalang.
- [0.5 ball] Parallel plastinkalar bir xil gravitatsion maydon (og‘irlik kuchi maydoni) g ga tik joylashgan. Dastlab tinch turgan diskni yuqoriga ko‘tarish uchun qo‘llanilgan kuchlanish V_{th} chegaraviy qiymatdan oshishi kerak.

V_{th} ni m , g , d va χ larda ifodalang.

- [2.3 ball] $V > V_{th}$ bo‘lganda, disk plastinkalar orasida yuqoriga-pastga harakat qiladi (disk faqat vertikal, tebranishsiz harakatlanadi deb faraz qiling). Disk va plastinkalar orasidagi to‘qnashuvlar noelastik (energiya saqlanmaydi), qayta tiklash koeffitsienti (to‘qnashuvdan keyingi va oldingi tezliklar nisbati) $\eta = V_{keyin}/V_{oldin}$ bilan berilgan, bu yerda V_{oldin} va V_{keyin} diskning to‘qnashuvdan oldingi va keyingi tezliklari. Plastinkalar joyida qattiq mahkamlangan. Diskning pastki plastinkada to‘qnashuvdan keyingi tezligi “turg‘un tezlik” v_s ga yaqinlashadi, u V ga quyidagicha bog‘liq:

$$v_s = \sqrt{\alpha V^2 + \beta}.$$

α va β koeffitsientlarini m , g , χ , d va η larda ifodalang. Diskning butun yuzasi plastinkaga tekis va bir vaqtda tegadi, shunda har bir to‘qnashuvda zaryad almashinuvi (disk va plastinka orasidagi zaryad o‘tishi) bir zumda ro‘y beradi deb faraz qiling.

- (e) [2.2 ball] Turg'un holatga erishgandan so'ng, kondensator plastinkalari orqali o'rtacha tok (vaqt bo'yicha o'rtalashtirilgan elektr toki) I ni $qV \gg mgd$ shartida $I = \gamma V^2$ deb yaqinlashtirish mumkin. γ koeffitsientini m , χ , d va η larda ifodalang.
- (f) [3 ball] Qo'llanilgan kuchlanish V (juda sekin) kamaytirilganda, kritik kuchlanish V_c mavjud bo'lib, undan pastda zaryad oqimi to'xtaydi. V_c va unga mos keluvchi tok I_c ni m , g , χ , d va η larda toping. V_c ni (c) banddagi ko'tarilish chegarasi V_{th} bilan solishtirib, V ni 0 dan $3V_{th}$ gacha oshirish va kamaytirishdagi $I-V$ xarakteristikasining (tok kuchlanishga bog'liqlik grafigi) taxminiy eskizini chizing.

2-nazariy savol: Ko'tariluvchi shar (yerdan yuqoriga ko'tariladigan rezina shar)

Rezina shar geliy gazi bilan to'ldirilgan va balandlikka ko'tariladi, bu yerda bosim va harorat balandlik bilan kamayadi. Quyidagi savollarda, sharning shakli yuklamadan qat'iy nazar sferik (sharsimon) bo'lib qoladi deb faraz qiling va yuklamaning hajmini e'tiborsiz qoldiring. Shuningdek, shar ichidagi geliy gazining harorati har doim atrofdagi havoning haroratiga teng deb faraz qiling va barcha gazlarni ideal gazlar (molekulalararo tortishish kuchlari e'tiborga olinmaydigan gazlar) deb hisoblang. Universal gaz doimiysi $R = 8.31 \text{ J/mol}\cdot\text{K}$, geliy va havoning molyar massalari mos ravishda $M_H = 4.00 \times 10^{-3} \text{ kg/mol}$ va $M_A = 28.9 \times 10^{-3} \text{ kg/mol}$. Gravitatsion tezlanish $g = 9.8 \text{ m/s}^2$.

A qism

- (a) [1.5 ball] Atrofdagi havoning bosimi P va harorati T bo'lsin. Shar ichidagi bosim shar sirt tarangligi (sirtning qisqarishga intilishi) tufayli tashqi bosimdan yuqori. Shar ichida n mol geliy gazi bor va ichki bosim $P + \Delta P$. Sharga ta'sir etuvchi suzuvchi kuchni (Arximed kuchi) F_B ni P va ΔP orqali ifodalang.
- (b) [2 ball] Koreyada yozning ma'lum bir kunida dengiz sathidan z balandlikdagi havo harorati $T(z) = T_0(1 - z/z_0)$ ko'rinishida ekani aniqlandi, bu $0 < z < 15 \text{ km}$ oralig'ida, $z_0 = 49 \text{ km}$ va $T_0 = 303 \text{ K}$ bilan. Dengiz sathidagi bosim va zichlik mos ravishda $P_0 = 1.0 \text{ atm} = 1.01 \times 10^5 \text{ Pa}$ va $\rho_0 = 1.16 \text{ kg/m}^3$. Ushbu balandlik oralig'ida bosim quyidagi ko'rinishga ega:

$$P(z) = P_0(1 - z/z_0)^\eta. \quad (2.1)$$

η ni z_0 , ρ_0 , P_0 va g larda ifodalang va uning son qiymatini ikkita muhim raqamgacha toping. Gravitatsion tezlanishni balandlikka bog'liq emas, o'zgarmas deb hisoblang.

B qism

Cho'zilmagan radiusi r_0 bo'lgan sferik rezina shar radiusi r ($\geq r_0$) gacha shishirilganda, shar sirtida cho'zilish tufayli qo'shimcha elastik energiya (deformatsiyada saqlanadigan energiya) mavjud. Soddalashtirilgan nazariyada, o'zgarmas harorat T dagi elastik energiya quyidagicha ifodalanadi:

$$U = 4\pi r_0^2 \kappa R T \left(2\lambda^2 + \frac{1}{\lambda^4} - 3 \right) \quad (2.2)$$

bu yerda $\lambda = r/r_0$ (≥ 1) – o'lchamning shishish nisbati, κ esa ozgarmas (birligi mol/m^2).

- (c) [2 ball] ΔP ni (2.2) tenglamada berilgan parametrlar orqali ifodalang va ΔP ning $\lambda = r/r_0$ ga bog'liqlik grafigining taxminiy eskizini chizing.

- (d) [1.5 ball] κ o'zgarmasini sharni shishirish uchun zarur bo'lgan gaz miqdoridan aniqlash mumkin. $T_0 = 303\text{ K}$ va $P_0 = 1.0\text{ atm} = 1.01 \times 10^5\text{ Pa}$ da, cho'zilmagan shar ($\lambda = 1$) tarkibida $n_0 = 12.5\text{ mol}$ geliy bor. Sharni xuddi shu T_0 va P_0 da $\lambda = 1.5$ gacha shishirish uchun jami $n = 3.6 \times n_0 = 45\text{ mol}$ gaz kerak. $a = \kappa/\kappa_0$ deb belgilangan shar parametrini $\kappa_0 = \frac{r_0 P_0}{4RT_0}$ orqali ifodalang va a ni n , n_0 , λ larda toping. a ning son qiymatini ikkita muhim raqamgacha hisoblang.

C qism

Bir shar (d) banddagi kabi dengiz sathida tayyorlanadi ($\lambda = 1.5$ gacha shishirilgan, $n = 3.6 \times n_0 = 45\text{ mol}$ geliy gazi $T_0 = 303\text{ K}$ va $P_0 = 1\text{ atm} = 1.01 \times 10^5\text{ Pa}$ da). Gaz, sharning o'zi va boshqa yuklamalar qo'shilgan umumiy massa $M_T = 1.12\text{ kg}$. Endi shar dengiz sathidan ko'tarilsin.

- (e) [3 ball] Aytaylik, shar nihoyat z_f balandlikda to'xtasin, bu yerda suzuvchi kuch umumiy og'irlikni muvozanatlaydi. z_f va shu balandlikdagi shishish nisbati λ_f ni toping. Javoblarni ikkita muhim raqamgacha bering. Ko'tarilish davomida hech qanday siljish (yon tomonga og'ish) va gaz sizib chiqishi yo'q deb faraz qiling.

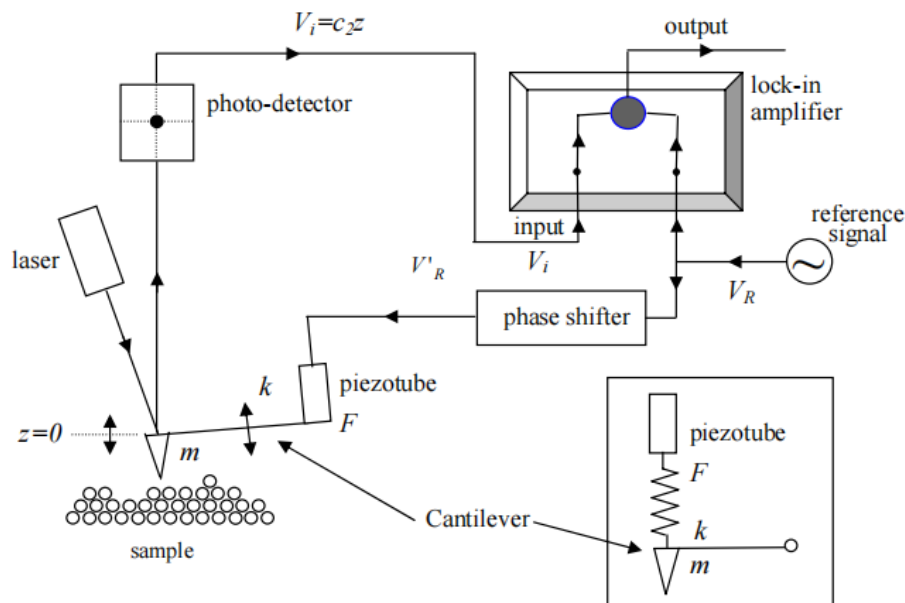
balandlik. Javoblarni ikkita muhim raqamgacha bering. Ko'tarilish davomida hech qanday siljish (yon tomonga og'ish) va gaz sizib chiqishi yo'q deb faraz qiling.

3-nazariy savol: Atom zond mikroskopi (APM – Atomic Probe Microscope)

Atom zond mikroskoplari nano-fan sohasida kuchli asboblardir. APM dagi konsolning (kantilever – yupqa elastik plastinka) harakati aks ettirilgan lazer nurini kuzatuvchi foto-detektor yordamida aniqlanishi mumkin, 3.1-rasmda ko'rsatilganidek. Konsol faqat vertikal yo'nalishda harakatlana oladi va uning z siljishi (joyidan ko'chishi) vaqt t ga quyidagi tenglama bilan bog'liq:

$$m \frac{d^2 z}{dt^2} + b \frac{dz}{dt} + kz = F, \quad (3.1)$$

bu yerda m – konsol massasi, $k = m\omega_0^2$ – konsolning prujina doimiysi (qattqlik koeffitsienti), b – kichik so'ndirish koeffitsienti (tebranishni susaytiruvchi kattalik), $\omega_0 \gg (b/m) > 0$ sharti bilan, va nihoyat F – piezoelektrik trubaning tashqi harakatlantiruvchi kuchi.



3.1-rasm: Skanerli zond mikroskopi (SPM) uchun sxematik chizma. Pastki o'ng burchakdagi kichik rasmda piezotrubaga bilan konsolning o'zaro ta'sirini tavsiflovchi soddalashtirilgan mexanik model ko'rsatilgan.

A qism

- (a) [1.5 ball] Agar $F = F_0 \sin \omega t$ bo'lsa, (3.1) tenglamani qanoatlantiruvchi $z(t)$ ni $z(t) = A \sin(\omega t - \phi)$ ko'rinishida yozish mumkin, bu yerda $A > 0$ va $0 \leq \phi \leq \pi$. A amplituda (tebranish eng katta siljishi) va $\tan \phi$ ifodalarini F_0, m, ω, ω_0 va b larda toping.

Amplituda A va $\tan \phi$ ifodalarini F_0, m, ω, ω_0 va b larda toping. Rezonans chastotasi $\omega = \omega_0$ dagi A va ϕ fazani oling.

- (b) [1 ball] 3.1-rasmda ko'rsatilgan lock-in kuchaytirgichi (sinxron detektor) kirish signalini $V_R = V_{R0} \sin \omega t$ lock-in mos yozuv signaliga ko'paytiradi va so'ngra ko'paytirilgan signalning faqat doimiy tok (dc) komponentasini o'tkazadi. Kirish signali $V_i = V_{i0} \sin(\omega_i t - \phi_i)$ berilgan deb faraz qiling. Bu yerda V_{R0}, V_{i0}, ω_i va ϕ_i – hammasi musbat berilgan o'zgarmaslar. Nolga teng bo'lmagan chiqish signali olish uchun ω (>0) ga qanday shart qo'yiladi? Shu chastotada nolga teng bo'lmagan doimiy tok (dc) chiqish signalining kattaligi ifodasini yozing.
- (c) [1.5 ball] Faza almashtirgich (phase shifter) orqali lock-in mos yozuv kuchlanishi $V_R = V_{R0} \sin \omega t$ o'zgarib $V'_R = V_{R0} \sin(\omega t + \pi/2)$ ga aylanadi. V'_R piezoelektrik trubaga qo'llanilib, konsolga $F = c_1 V'_R$ kuchini beradi. So'ngra foto-detektor konsolning z siljishini $V_i = c_2 z$ kuchlanishiga aylantiradi. Bu yerda c_1 va c_2 – o'zgarmaslar. $\omega = \omega_0$ da doimiy tok (dc) chiqish signalining kattaligi ifodasini toping.
- (d) [2 ball] Konsol massasining kichik o'zgarishi Δm rezonans chastotasini $\Delta \omega_0$ ga siljitadi. Natijada, dastlabki rezonans chastotasi ω_0 dagi ϕ faza $\Delta \phi$ ga o'zgaradi. Faza siljishi $\Delta \phi = \pi/1800$ ga mos keladigan massa o'zgarishi Δm ni toping (bu fazani o'lchashdagi tipik aniqlik). Konsolning fizik parametrlari: $m = 1.0 \times 10^{-12}$ kg, $k = 1.0$ N/m va $(b/m) = 1.0 \times 10^3$ s $^{-1}$. $(1+x)^a \approx 1+ax$ va $\tan(\pi/2+x) \approx -1/x$ ($|x| \ll 1$) yaqinlashuvlaridan foydalaning.

B qism

Endi A qismda muhokama qilingan harakatlantiruvchi kuchdan tashqari, namuna (sample) tufayli konsolga ta'sir etuvchi ba'zi kuchlar mavjud bo'lgan vaziyatni ko'rib chiqamiz, 3.1-rasmda ko'rsatilganidek.

- (e) [1.5 ball] Qo'shimcha $f(h)$ kuchi faqat konsol bilan namuna yuzasi orasidagi h masofaga bog'liq deb faraz qilib, yangi muvozanat holati h_0 ni topish mumkin. $h = h_0$ atrofida $f(h) \approx f(h_0) + c_3(h - h_0)$ yozish mumkin, bu yerda $c_3 - h$ ga bog'liq bo'lmagan o'zgarmas. Yangi rezonans chastotasi ω'_0 ni ω_0 , m va c_3 larda ifodalang.
- (f) [2.5 ball] Namuna yuzasini gorizontal harakatlantirib skaner qilish vaqtida, $Q = 6e$ zaryadlangan konsol uchi sirt ostida biror masofada tuzoqqa tushgan (fazoda lokalizatsiyalangan) $q = e$ zaryadli elektronga duch keladi.

(fazoning ma'lum bir nuqtasida joylashgan) sirt ostida biror masofada tuzoqqa tushgan elektronga duch keladi. Elektron atrofida skaner qilganda, rezonans chastotasining maksimal siljishi $\Delta\omega_0 (= \omega'_0 - \omega_0)$ ω_0 dan ancha kichik ekani kuzatiladi. Maksimal siljishdagi konsoldan tuzoqqa tushgan elektrongacha bo'lgan d_0 masofani m , q , Q , ω_0 , $\Delta\omega_0$ va Kulon doimiysi k_e larda ifodalang. $\Delta\omega_0 = 20 \text{ s}^{-1}$ uchun d_0 ni nm ($1 \text{ nm} = 1 \times 10^{-9} \text{ m}$) da hisoblang.

Konsolning fizik parametrlari: $m = 1.0 \times 10^{-12} \text{ kg}$ va $k = 1.0 \text{ N/m}$. Konsol uchida ham, sirtida ham hech qanday qutblanish effektini (dielektrikda zaryadlarning joylashishini) e'tiborga olmang. $k_e = 1/4\pi\epsilon_0 = 9.0 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$ va $e = -1.6 \times 10^{-19} \text{ C}$ ekanligini eslatib o'tamiz.

Eksperiment uchun ko'rsatmalar

Iltimos, quyidagi ko'rsatmalarni diqqat bilan o'qing:

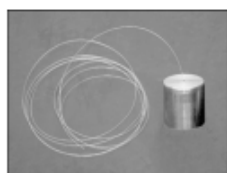
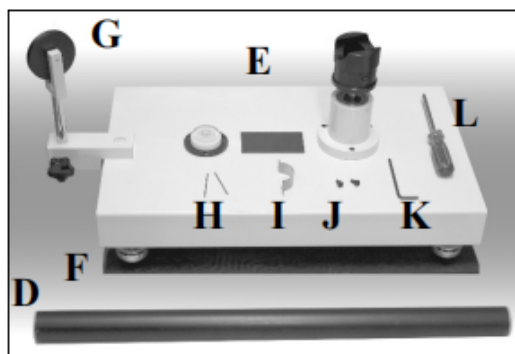
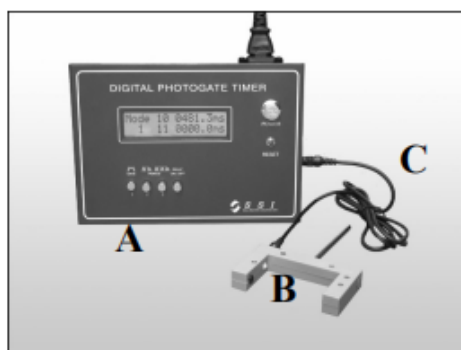
1. Mavjud vaqt 5 soat.
2. Faqat berilgan qalamdan foydalaning.
3. Yozuv qog'ozlarining faqat old tomonidan foydalaning. Faqat belgilangan maydon ichiga yozing.
4. Bo'sh yozuv qog'ozlaridan tashqari, olingan natijalarni qisqacha yozish uchun "Javob shakllari" (Answer Forms) mavjud.
5. O'lchovlaringizning natijalarini va savol yechimi uchun zarur deb hisoblagan barcha ma'lumotlarni bo'sh yozuv qog'ozlariga yozing. Iltimos, matndan imkon qadar kam foydalaning; asosan tenglamalar, raqamlar, chizmalar va grafiklar bilan ifodalang.
6. Har bir qog'oz varag'ining yuqori qismidagi katakchalarga mamlakat kodingizni (Country Code) va o'quvchi raqamingizni (Student Code) yozing. Bundan tashqari, har bir bo'sh yozuv varag'iga varaqning tartib raqamini (Page Number) va ishlatilgan yozuv varaqlarining umumiy sonini (Total Number of Pages) yozing. Agar ba'zi bir bo'sh yozuv varaqlarini baholanishini istamagan yozuvlar uchun ishlatasangiz, butun varaq bo'ylab katta X belgisini qo'ying va uni raqamlashingizga kiritmang.
7. Tajriba oxirida barcha varaqlarni quyidagi tartibda joylashtiring:
 - (a) Javob shakllari (Answer forms) (eng tepada)
 - (b) Ishlatilgan yozuv varaqlari tartib bilan

- (c) Baholanishini istamagan varaqlar
 - (d) Ishlatilmagan yozuv varaqlari
 - (e) Chop etilgan savol (eng pastda)
8. Olingan qiymatlaringizning xatolik oralig'ini ko'rsatish shart emas. Biroq ularning haqiqiy qiymatlardan farqlanishi sizning balingizni aniqlaydi.
9. Qog'ozlarni konvert ichiga soling va hamma narsani stolingizda qoldiring. Tajribada ishlatilgan hech qanday qog'oz yoki materialni xonadan olib chiqishga ruxsat yo'q.

Jihozlar va materiallar

Nomi	Miqdori	Nomi	Miqdori
Fotodarvoza taymeri (Photogate timer)	1	Philips tornavida	1
Fotodarvoza (Photogate)	1	Ip bilan og'irlik	1
Ulagich kabeli	1	Elektron tarozi	1
Mexanik "qora quti" (qora silindr)	1	Chizg'ichli stend	1
Aylanma stol (Rotation stage)	1	U-shaklidagi tayanch	1
Rezina to'shak	1	C-qisqich	1
Shkiv (Pulley)	2	Chizg'ich (0.50 m, 0.15 m)	har biri 1
Pin (tirgak)	2	Noniusli shtangensirkul	1
U-shaklidagi plastinka	1	Qaychi	1
Vint	2	Ip (thread)	1
Allen kaliti (olti qirrali, L-shaklidagi)	1	Ehtiyot qismlar (ip, tirgak, vint, Allen kaliti)	1

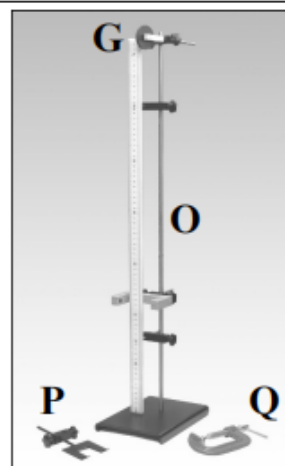
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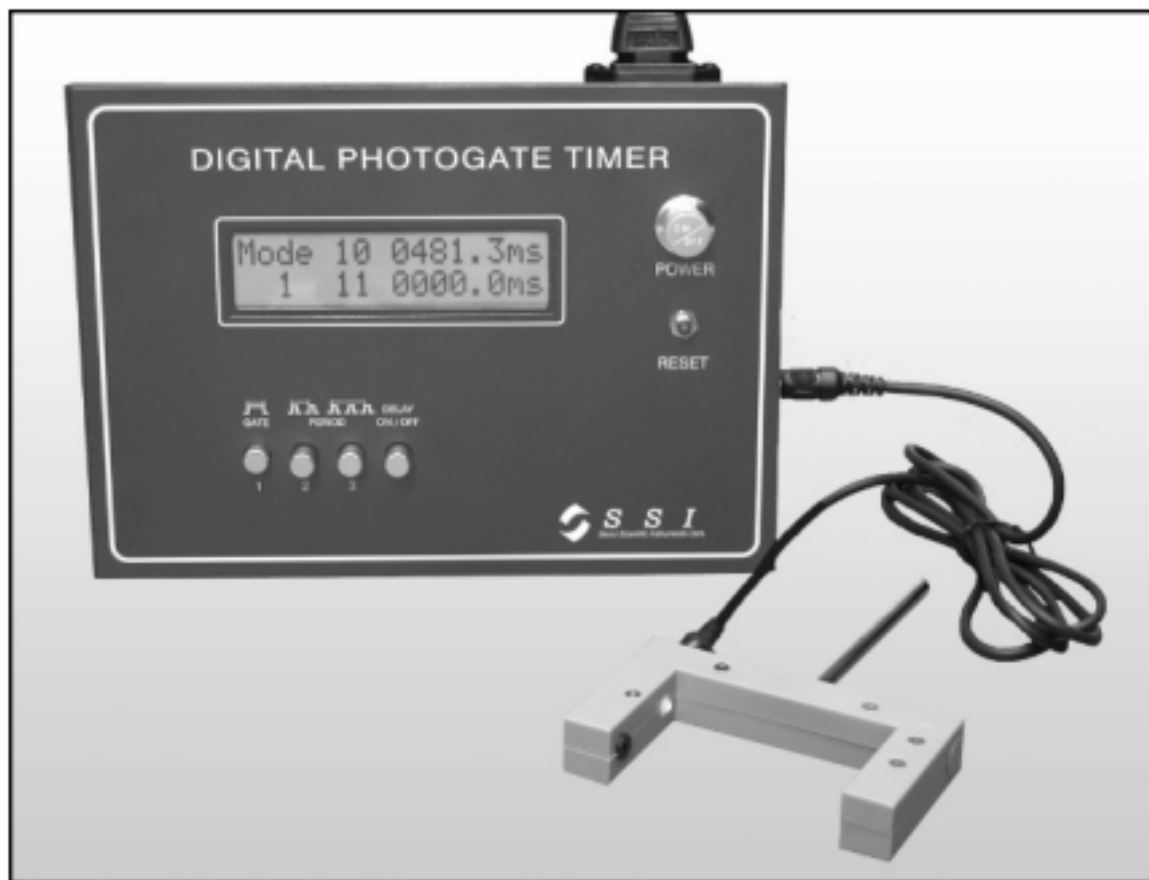


Fotodarvoza taymeri uchun korsatma (Photogate Timer – yoruglik tosigi taymeri)

Fotodarvoza (Photogate) infraqizil LED va fotodetektordan iborat. Fotodarvoza taymeriga ulab, datchikka yetib boruvchi infraqizil nurning tosilishi bilan bogliq vaqt oraligini olchash mumkin.

- Fotodarvoza taymarga ulanganligiga ishonch hosil qiling. “POWER” tugmasini bosib quvvatni yoqing.
- Bitta tosilish hodisasining vaqt davomiyligini olchash uchun “GATE” (DARVOZA) tugmasini bosing. Tezlik olchashlari uchun ushbu “GATE” rejimidan foydalaning.
- Ikki yoki uchta ketma-ket tosilish hodisalari orasidagi vaqt intervalini olchash uchun mos “PERIOD” (DAVR) tugmasini bosing. Tebranish olchashlari uchun ushbu “PERIOD” rejimidan foydalaning.
- Agar “DELAY” (KECHIKTIRISH) tugmasi bosilgan (ichkariga) bolsa, fotodarvoza taymeri har bir olchov natijasini 5 soniya davomida korsatadi va keyin ozini ozi tiklaydi (reset).
- Agar “DELAY” tugmasi chiqarilgan (tashqariga) bolsa, fotodarvoza taymeri keyingi olchov tugaguniga qadar oldingi olchov natijasini korsatadi.
- Tugma holatidagi har qanday ozgarishdan song, rejim ozgarishini faollashtirish uchun “RESET” (TIKLASH) tugmasini bir marta bosing.

Ehtiyot: Fotodarzovaga togridan-togri qaramang. Korinmas infraqizil nur kozlarga zarar yetkazishi mumkin.



Photogate, Photogate Timer, and connection cable

Fotodarvoza, fotodarvoza taymeri va ulanish kabeli

Elektron tarozi uchun korsatma (Electronic Balance – elektron tarozi)

- Tarozi barqaror qilish uchun pastki oyoqlarini sozlang. (Sath korsatkichi (level indicator) mavjud bolsa ham, tarozini toliq gorizontol holatga keltirish shart emas.)
- Taroziga hech narsa qoymasdan, “On/Off” (YOQ/OCHIR) tugmasini bosib uni yoqing.
- Dumaloq tortish panasiga (weighing pan) jismni qoying. Uning massasi grammlarda korsatiladi.
- Agar tortish panasida hech narsa bolmasa, tarozi taxminan 25 soniyada avtomatik ravishda ochadi.



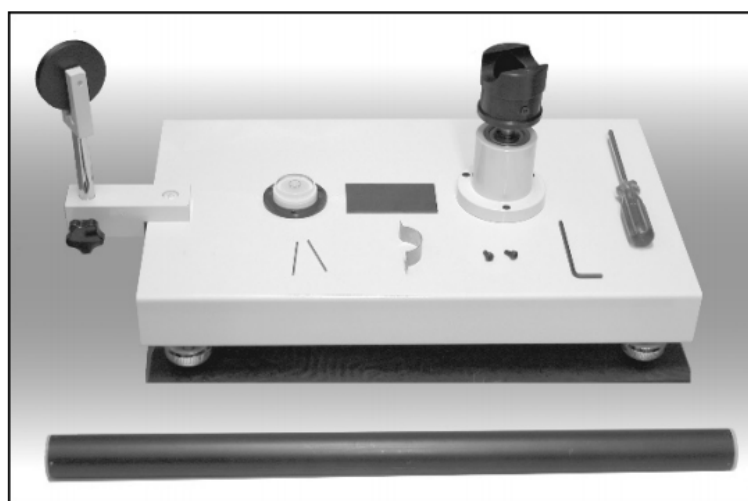
Balance

Tarozi (Balance)

Aylanma stol (rotation stage) uchun korsatma

- Aylanma stolni rezina toshak (rubber pad) ustida, deyarli gorizontaal holatda barqaror turishi uchun pastki oyoqlarini sozlang.
- U-shaklidagi plastinka va ikkita vint yordamida mexanik “Qora quti”ni (Mechanical “Black Box” – qora silindr) aylanuvchi stol ustuni (rotating stub) tepasiga ornating. Vintlarni mahkamlash uchun Allen kalitidan (olti qirrali, L-shaklidagi) foydalaning.
- Ogirlikka boglangan ip (string) aylanuvchi stol ustuni yonidagi vintga mahkamlanadi. Buning uchun Philips tornavidadan foydalaning.

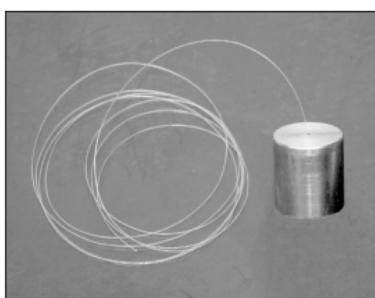
Ehtiyot: Mexanik “Qora quti” aylanayotgan vaqtda unga juda yaqin qaramang. Kozlaringiz shikastlanishi mumkin.



Mechanical “Black Box” and rotation stage



Rotating stub



Weight with a string

Mexanik “Qora quti” va aylanma stol tasviri

Aylanuvchi stol ustuni (rotating stub)

Ipli ogirlik (weight with a string)

Mexanik “Qora quti” (Mechanical “Black Box”)

Savol: “Qora quti” ichidagi sharning massasini va ikkita prujinaning prujina doimiylarini (spring constant – qattiqlik koeffitsienti) toping.

Mexanik “Qora quti” haqida umumiy malumot

Mexanik “Qora quti” (MBB) qora silindrsimon trubka ichida ikkita prujinaga mahkamlangan qattiq shardan iborat, 1-rasmda korsatilganidek. Ikki prujina bir xil mahkam oralgan simdan tayyorlangan, faqat oramlar soni (number of turns) har xil. Prujinalarning massalari va chozil-magan holatdagi uzunliklari etiborga olinmasin. Trubka bir jinsli (homogeneous) va ikkita bir xil uchlik qopqoq (end cap) bilan muhrlangan. Qopqoqlarning trubka ichiga kiruvchi qismining uzunligi 5 mm. Sharining radiusi 11 mm va trubkaning ichki diametri 23 mm. Gravitatsion tezlanish $g = 9.8 \text{ m/s}^2$ deb berilgan. Shar va trubkaning ichki devorlari orasida chekli ishqalanish (finite friction) mavjud.

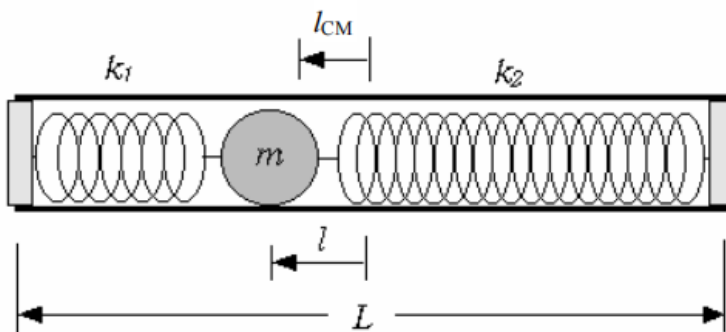


Fig 1 Mechanical “Black Box” (not to scale)

1-rasm: Mexanik “Qora quti” (masshtabga rioya qilinmagan)

Ushbu eksperimentning maqsadi MBB ni ochmasdan, sharining massasi m va prujinalarning k_1 va k_2 prujina doimiylarini topishdir. Muammoning qiyin tomoni shundaki, hech qanday bitta tajriba m massani yoki sharining l joylashuvini (position) bera olmaydi, chunki bu ikki kattalik bir-biri bilan boglangan. Bu yerda l – MBB gorizontol holatda muvozanatda (ishqalanish nol bolganda) turgan vaqtdagi trubka markazi bilan shar markazi orasidagi masofa.

Quyida keltirilgan belgilar qiziqish uygotuvchi fizik kattaliklarni ifodalash uchun ishlatilishi kerak. Agar boshqa fizik kattaliklardan foydalanish zarur bolsa, chalkashlikka yol qoymaslik uchun quyida berilgan belgilardan farqli belgilardan foydalaning.

Belgilangan fizik belgilar (Assigned Physical Symbols)

Sharning massasi (mass of the ball):	m
Sharning radiusi (radius of the ball):	$r(= 11 \text{ mm})$
MBB ning shardan tashqari qismining massasi:	M
Qora trubkaning uzunligi:	L
Trubka ichiga kiruvchi har bir qopqoq uzunligi:	$\delta(= 5.0 \text{ mm})$
MBB ning massa markazidan trubka markazigacha bolgan masofa:	l_{CM}
Shar markazi bilan trubka markazi orasidagi masofa:	x (yoki MBB gorizontal holatda m)
Gravitatsion tezlanish:	$g(= 9.8 \text{ m/s}^2)$
Ipga boglangan ogirlikning massasi:	m_0
Ogirlikning tezligi:	v
Ogirlikning pastga siljishi:	h
Ip oraladigan aylanuvchi stol ustunining radiusi:	R
Inersiya momentlari (moments of inertia):	I, I_0, I_1, I_2 , va hokazo
Burchak tezlik va burchak chastotalari:	$\omega, \omega_1, \omega_2$, va hokazo
Tebranish davrlari (periods of oscillation):	T_1, T_2
Effektiv umumiy prujina doimiysi:	k
Ikki prujinaning prujina doimiylari:	k_1, k_2
Prujinalarning oramlar soni:	N_1, N_2

Ehtiyot: MBB ni ochishga urinmang. Agar ochsangiz, siz diskvalifikatsiya qilinasiz va Eksperimental musobaqadagi balingiz nolga teng boladi.

Ehtiyot: MBB ni kuchli silkitmang va yerga tushirmang. Shar prujinalardan ajralib ketishi mumkin. Agar MBB yaroqsiz bolib qolgandek tuyulsa, darhol proktorlarga xabar bering. U faqat bir marta balingizga tasir qilmasdan almashtiriladi. Har qanday keyingi almashtirish har safar balingizni 0.5 ballga kamaytiradi.

A-QISM: Sharning massasi va joylashuvi kopaytmasi ($m \times l$) (4.0 ball)

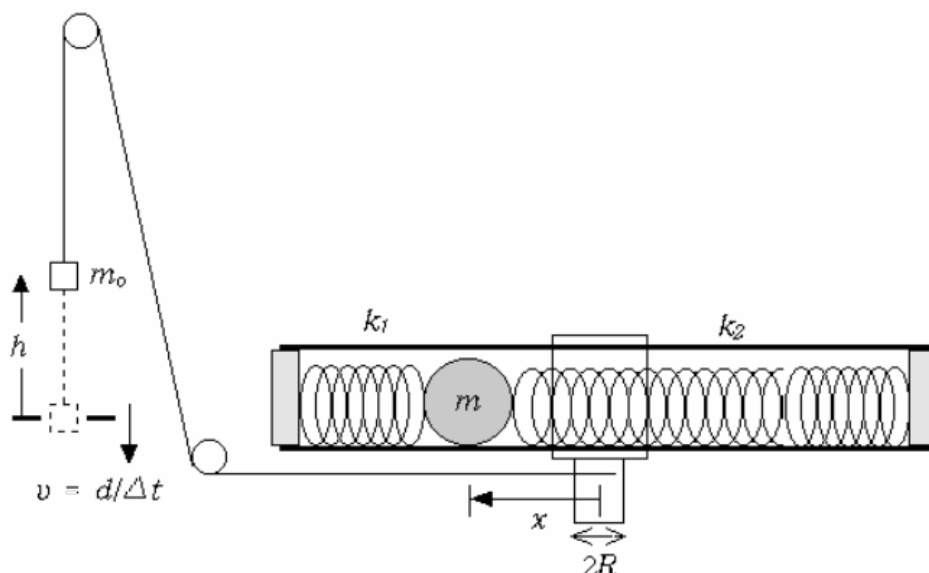
l – MBB gorizontal holatda muvozanatda turganida (1-rasmdagidek) shar markazining trubka markaziga nisbatan joylashuvi. Eksperimental ravishda m massa va l joylashuv kopaytmasining qiymatini toping. Buning natijasidan B-QISMda m qiymatini aniqlashda foydalanasiz.

1. $m \times l$ ni olish imkonini beruvchi usulni tenglamalar yordamida taklif qiling va asoslang. (2.0 ball)
2. $m \times l$ ning qiymatini eksperimental ravishda aniqlang. (2.0 ball)

B-QISM: Sharning massasi m (10.0 ball)

2-rasmda aylanuvchi stol ustuniga gorizontal ravishda mahkamlangan MBB va bir uchi aylanuvchi stol ustuniga oralgan ipning ikkinchi uchiga boglangan ogirlik korsatilgan. Ogirlik tushganda ip ochiladi va MBB aylanadi. Ushbu tajribaga tegishli tenglamani A-QISMda olingan tenglama bilan birlashtirib, m uchun tenglama topishingiz mumkin.

Silindrsimon trubkaning ichki devorlari bilan shar orasida ishqalanish kuchi (frictional force) ta'sir qiladi. Ishqalanishning fizik mexanizmlari va aylanma harakat ostida shar sirpanishining tafsilotlari murakkab. Tahlilni soddalashtirish uchun kinetik ishqalanish (sirpanish ishqalanishi) tufayli energiya tarqalishini e'tiborsiz qoldirishingiz mumkin.



2-rasm: Mexanik “Qora quti” ning aylanishi (masshtabga rioya qilinmagan). MBB ning burchak tezligi ω ni fotodarvoza orqali o'tayotgan ogirlik tezligi v dan olish mumkin. x – sharning aylanish oqiga nisbatan joylashuvi, d – ogirlikning uzunligi.

1. Ogirlikning turli tushish masofalari h uchun ogirlik tezligi v ni olchang. $h = 1.0 \times 10^{-2}$ m dan 4.0×10^{-1} m gacha bolgan butun diapazonni har bir h da v ni bir marta olchab, $1.0 \times 10^{-2} \sim 2.0 \times 10^{-2}$ m oraliqda skanerlash tavsiya etiladi. Ma'lumotlarni grafik qogoziga m qiymatini topishga qulay shaklda chizing. v va h orasidagi bogliqlik haqida umumiy tasavvurga ega bolgach, agar kerak bolsa, olchovni takrorlashingiz yoki bir necha nuqta qoshishingiz mumkin. MBB sekin aylanganda, shar va trubka orasidagi ishqalanish tufayli shar o'zining statik muvozanat holatidan sirpanmaydi. MBB yetarlicha tez aylanganda, shar prujinalar kuchsiz bolgani uchun qopqoqqa uriladi va haqiqatda qopqoqda qoladi. Grafikda sekin aylanish mintaqasini va tez aylanish mintaqasini aniqlang. (4.0 ball)
2. Olchovlaringiz sekin aylanish mintaqasida h ning v^2 ga proporsional ($h = Cv^2$) ekanligiga mos kelishini korsating. Olchovlaringizdan tez aylanish mintaqasida $h = Av^2 + B$ ekanligini korsating. (1.0 ball)
3. Radiusi r va massasi m bolgan sharning oz markazidan otuvchi oqqa nisbatan inersiya momenti (moment of inertia) $2mr^2/5$ ga teng. Agar shar oqqa perpendikulyar ravishda a masofaga siljigan bolsa, inersiya momenti ma^2 ga ortadi. Shar bilan birga aylanuvchi barcha jismlarning (shardan tashqari) umumiy inersiya momentini ifodalash uchun I belgisidan foydalaning. C koeffitsientini MBB ning m , l kabi parametrlari bilan boglang. (1.0 ball)
4. A va B koeffitsientlarini MBB ning m , l kabi parametrlari bilan boglang. (1.0 ball)
5. Olchovlaringiz va A-QISMda olingan natijalardan foydalanib, m ning qiymatini aniqlang. (3.0 ball)

C-QISM: Prujina doimiylari k_1 va k_2 (6.0 ball)

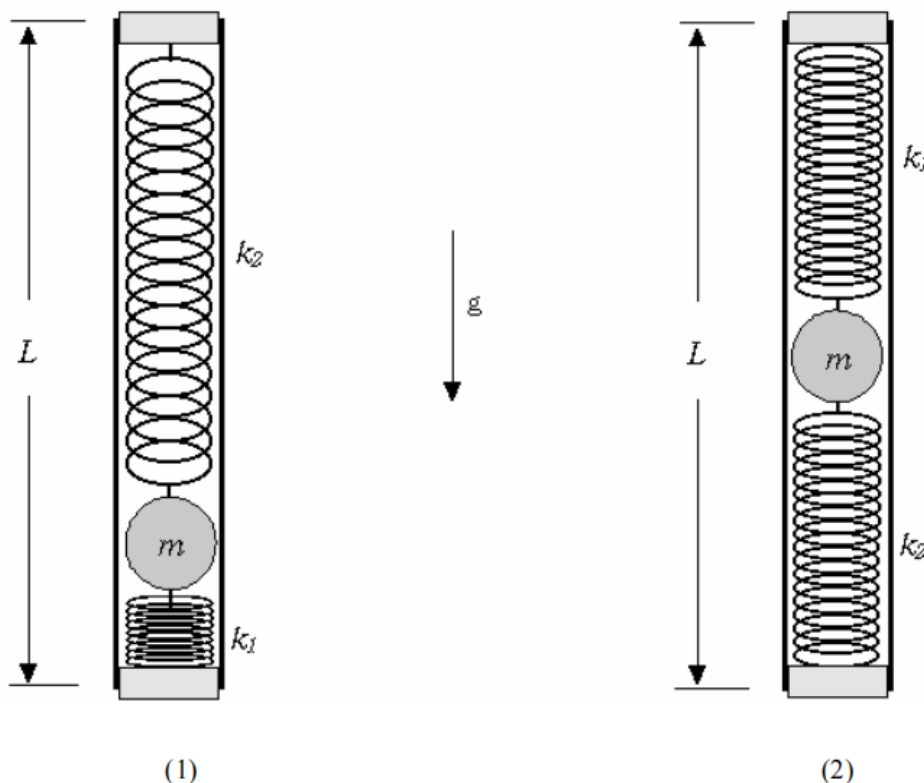
Ushbu qismda siz MBB ni qattiq mayatnik (rigid pendulum) sifatida ishlatib, kichik tebranish tajribalarini bajarishingiz kerak. MBB ning har ikki uchida ikkita kichik teshik mavjud. Bu teshiklarga kiritilgan ikkita yupqa pin (tirgak) kichik tebranishning aylanish oqi (pivot) sifatida

ishlatilishi mumkin. U-shaklidagi tayanch stendga qisqich bilan mahkamlanadi va aylanish oqini ushlab turish uchun ishlatiladi.

Kichik tebranishning burchak chastotasi ω quyidagicha berilishini unutmang:

$$\omega = \left[\frac{\text{aylantiruvchi moment (torque)}}{\text{inersiya momenti} \times \text{burchak (angle)}} \right]^{1/2}.$$

Bu yerda aylantiruvchi moment va inersiya momenti aylanish oqiga nisbatan olinadi. B-QISMga oxshab, shardan tashqari MBB ning noma'lum inersiya momenti I_0 dan qochish uchun 3-rasmda korsatilgan ikkita eksperimental shartni korib chiqing.



3-rasm: Mexanik “Qora quti” ning tebranishi (masshtabga rioya qilinmagan). Yuqorida korsatilgan ikki konfiguratsiya uchun kichik tebranish davrlari T_1 va T_2 fotodarvoza yordamida olchanishi mumkin. Ushbu tajriba uchun ikkita pin va U-shaklidagi tayanch taqdim etilgan.

1. 3-rasm (1) va (2) da korsatilgan kichik tebranish davrlari T_1 va T_2 ni olchang va ularning qiymatlarini yozing. (1.0 ball)
2. Nima uchun konfiguratsiyalarning kichik tebranish burchak chastotalari ω_1 va ω_2 har xil ekanligini (tenglamalar yordamida) tushuntiring. MBB ning shardan tashqari qismining uchidagi MBB ga perpendikulyar oqqa nisbatan inersiya momentini ifodalash uchun I_0 belgisidan foydalaning. Sharhning gorizontol muvozanat holatidan siljishini ifodalash uchun Δl belgisidan foydalaning. (1.0 ball)
3. I_0 ni oldingi natijalardan yoqotib, Δl ni hisoblang. (1.0 ball)
4. C-QISM 1–3 va B-QISM natijalarini birlashtirib, ikki prujinali tizimning effektiv umumiy prujina doimiysi k ning qiymatini toping va yozing. (2.0 ball)
5. k_1 va k_2 ning mos qiymatlarini oling. Ularning qiymatlarini yozing. (1.0 ball)

1-nazariy savol: Ping-Pong rezistor (ping-pong rezistori)

1. Javoblar

(a)

$$F_R = -\frac{1}{2}\pi R^2 \varepsilon_0 \frac{V^2}{d^2}$$

(b)

$$\chi = -\varepsilon_0 \frac{\pi r^2}{d}$$

(c)

$$V_{\text{th}} = \sqrt{\frac{2mgd}{\chi}}$$

(d)

$$v_s = \sqrt{\alpha V^2 + \beta}$$

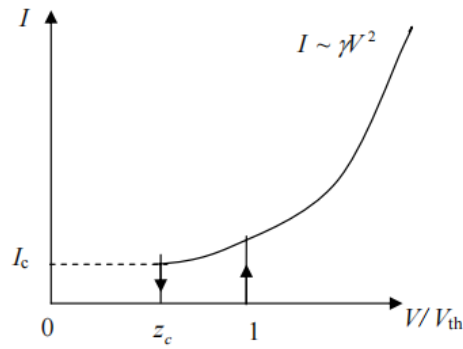
$$\alpha = \left(\frac{\eta^2}{1 - \eta^2} \right) \left(\frac{2\chi}{m} \right), \quad \beta = \left(\frac{\eta^2}{1 + \eta^2} \right) (2gd)$$

(e)

$$\gamma = \sqrt{\frac{1 + \eta}{1 - \eta}} \sqrt{\frac{\chi^3}{2md^2}}$$

(f)

$$V_c = \sqrt{\frac{1 - \eta^2}{1 + \eta^2}} \sqrt{\frac{mgd}{\chi}}, \quad I_c = \frac{2\eta\sqrt{1 - \eta^2}}{(1 + \eta)(1 + \eta^2)} g\sqrt{m\chi}$$



I-V xarakteristikasi (taxminiy)

2. Yechimlar

(a) [1.2 ball]

Tashqi kuchlanish V ta'sirida hosil bo'lgan Q zaryadni Gauss qonuni yordamida topish mumkin:

$$\varepsilon_0 \oint \vec{E} \cdot d\vec{s} = Q \quad (\text{a1})$$

$$Q = \varepsilon_0 E \cdot (\pi R^2) = \varepsilon_0 \left(\frac{V}{d} \right) \cdot (\pi R^2), \quad (\text{a2})$$

bu yerda $V = Ed$.

Kondensatorida saqlanuvchi energiya:

$$U = \int_0^V Q(V') dV' = \int_0^V \varepsilon_0 \pi R^2 \left(\frac{V'}{d} \right) dV' = \frac{1}{2} \varepsilon_0 \pi R^2 \frac{V^2}{d}. \quad (\text{a3})$$

Plastinkaga ta'sir etuvchi kuch (V o'zgarmas):

$$\therefore F_R = \pm \frac{\partial U}{\partial d} = -\frac{1}{2} \varepsilon_0 \pi R^2 \frac{V^2}{d^2}. \quad (\text{a4})$$

[Alternativ yechim:] Bir plastinkaga ta'sir qiluvchi elektr maydon E' faqat ikkinchi plastinka tomonidan hosil qilinadi va uning kattaligi

$$E' = \frac{1}{2} E = \frac{V}{2d}, \quad (\text{a5})$$

shuning uchun plastinkaga ta'sir etuvchi kuch:

$$F_R = QE'. \quad (\text{a6})$$

(b) [0.8 ball]

Kichik diskdagi q zaryadini ham Gauss qonuni yordamida hisoblash mumkin:

$$\varepsilon_0 \oint \vec{E} \cdot d\vec{s} = q. \quad (\text{b1})$$

Diskning bir tomoni plastinkaga tegib turganligi sababli,

$$q = -\varepsilon_0 E \cdot (\pi r^2) = -\varepsilon_0 \frac{\pi r^2}{d} V = \chi V. \quad (\text{b2})$$

Yoki maydon nisbatidan: $q = \left(\frac{\pi r^2}{\pi R^2}\right) Q$.

$$\therefore \chi = -\varepsilon_0 \frac{\pi r^2}{d} \quad (\text{b3})$$

(c) [0.5 ball]

Kichik diskka ta'sir etuvchi aniq kuch F_{net} gravitatsion va elektrostatik kuchlarning yig'indisidan iborat:

$$F_{\text{net}} = F_g + F_e \quad (\text{c1})$$

Gravitatsion kuch: $F_g = -mg$.

Elektrostatik kuchni (a) band natijasidan olish mumkin:

$$F_e = \frac{1}{2} \varepsilon_0 \frac{\pi r^2}{d^2} V^2 = \frac{\chi}{2d} V^2 \quad (\text{c2})$$

Diskni ko'tarish uchun $F_{\text{net}} > 0$ bo'lishi kerak:

$$\frac{\chi}{2d} V^2 - mg > 0 \quad (\text{c3})$$

$$\therefore V_{\text{th}} = \sqrt{\frac{2mgd}{\chi}} \quad (\text{c4})$$

(d) [2.3 ball]

v_s – kichik diskning pastki plastinkaga to'qnashgandan keyingi turg'un tezligi bo'lsin. U holda pastki plastinka ustidagi diskning turg'un kinetik energiyasi:

$$K_s = \frac{1}{2} m v_s^2 \quad (\text{d1})$$

Har bir to'liq aylanishda disk elektrostatik energiya oladi:

$$\Delta U = 2qV \quad (\text{d2})$$

Har bir noelastik to'qnashuvda disk kinetik energiyasining yo'qotishi:

$$\Delta K_{\text{loss}} = K_{\text{before}} - K_{\text{after}} = (1 - \eta^2) K_{\text{before}} = \left(\frac{1}{\eta^2} - 1\right) K_{\text{after}} \quad (\text{d3})$$

K_s – pastki plastinkadagi to'qnashuvdan keyingi energiya va $(K_s + qV - mgd)$ – yuqori plastinkadagi to'qnashuvdan oldingi energiya bo'lgani uchun, to'liq aylanish davomidagi umumiy energiya yo'qotilishini K_s orqali yozish mumkin:

$$\Delta K_{\text{tot}} = \left(\frac{1}{\eta^2} - 1\right) K_s + (1 - \eta^2)(K_s + qV - mgd). \quad (\text{d4})$$

Turg'un holatda ΔU energiya ΔK_{tot} ga teng bo'lishi kerak:

$$2qV = \left(\frac{1}{\eta^2} - 1\right) K_s + (1 - \eta^2)(K_s + qV - mgd). \quad (\text{d5})$$

(d5) tenglamani tartibga keltirib:

$$\begin{aligned} K_s &= \frac{\eta^2}{1 - \eta^2} [(1 + \eta^2)qV + (1 - \eta^2)mgd] \\ &= \left(\frac{\eta^2}{1 - \eta^2}\right) qV + \left(\frac{\eta^2}{1 + \eta^2}\right) mgd \\ &= \frac{1}{2}mv_s^2. \end{aligned} \quad (\text{d6})$$

Shunday qilib,

$$v_s = \sqrt{\left(\frac{\eta^2}{1 - \eta^2}\right) \left(\frac{2\chi V^2}{m}\right) + \left(\frac{\eta^2}{1 + \eta^2}\right) (2gd)}. \quad (\text{d7})$$

Berilgan $v_s = \sqrt{\alpha V^2 + \beta}$ bilan solishtirsak:

$$\alpha = \left(\frac{\eta^2}{1 - \eta^2}\right) \left(\frac{2\chi}{m}\right), \quad \beta = \left(\frac{\eta^2}{1 + \eta^2}\right) (2gd). \quad (\text{d9})$$

[Alternativ yechim:] v_n – kichik diskning pastki plastinka bilan n -to'qnashuvdan keyingi tezligi bo'lsin. Pastki plastinka ustidagi kinetik energiya:

$$K_n = \frac{1}{2}mv_n^2. \quad (\text{d10})$$

Yuqori plastinkaga yetganda disk potensial energiyasining o'zgarishi hisobiga energiya oladi:

$$\Delta U_{\text{up}} = qV - mgd. \quad (\text{d11})$$

Shuning uchun yuqori plastinkaga to'qnashuv oldidan kinetik energiya:

$$K_{n-\text{up}} = \frac{1}{2}mv_n^2 = K_n + \Delta U_{\text{up}}. \quad (\text{d12})$$

$\eta = v_{\text{after}}/v_{\text{before}}$ bo'lgani uchun, yuqori plastinka bilan to'qnashgandan keyingi kinetik energiya η^2 marta kichiklashadi:

$$K'_{n \rightarrow \text{up}} = \eta^2 \cdot K_{n \rightarrow \text{up}} \quad (\text{d13})$$

Pastga harakatdagi potensial energiya ortishi:

$$\Delta U_{\text{down}} = qV + mgd \quad (\text{d14})$$

Shunday qilib, pastki plastinkaga to‘qnashuv oldidan kinetik energiya:

$$K_{n \rightarrow \text{down}} = K'_{n \rightarrow \text{up}} + \Delta U_{\text{down}} \quad (\text{d15})$$

Yana pastki plastinkadagi to‘qnashuvda energiya yo‘qolishi hisobga olinib, $(n+1)$ -to‘qnashuvdan keyingi kinetik energiya:

$$\begin{aligned} K_{n+1} &= \eta^2 \cdot K_{n \rightarrow \text{down}} \\ &= \eta^2 (K'_{n \rightarrow \text{up}} + \Delta U_{\text{down}}) \\ &= \eta^2 (\eta^2 (K_n + \Delta U_{\text{up}}) + \Delta U_{\text{down}}) \\ &= \eta^2 (\eta^2 (K_n + qV - mgd) + qV + mgd) \\ &= \eta^4 K_n + \eta^2 (1 + \eta^2) qV + \eta^2 (1 - \eta^2) mgd \end{aligned} \quad (\text{d16})$$

$n \rightarrow \infty$ da $v_n \rightarrow v_s$, ya'ni $K_n \rightarrow K_s = \frac{1}{2}mv_s^2$:

$$\begin{aligned} K_s &= \frac{1}{1 - \eta^4} [\eta^2 (1 + \eta^2) qV + \eta^2 (1 - \eta^2) mgd] \\ &= \left(\frac{\eta^2}{1 - \eta^2} \right) qV + \left(\frac{\eta^2}{1 + \eta^2} \right) mgd = \frac{1}{2}mv_s^2 \end{aligned} \quad (\text{d17})$$

(e) [2.2 ball]

Diskning plastinkalar orasidagi bir aylanishida olib o‘tgan zaryadi $\Delta Q = 2q$, va vaqt oralig‘i $\Delta t = t_+ + t_-$, bu yerda t_+ (t_-) – yuqoriga (pastga) harakat vaqti.

$$v_{0+}t_+ + \frac{1}{2}a_+t_+^2 = d \quad (\text{e1})$$

$$v_{0-}t_- + \frac{1}{2}a_-t_-^2 = d \quad (\text{e2})$$

bu yerda v_{0+} (v_{0-}) – pastki (yuqori) plastinkadagi boshlang‘ich tezlik, a_+ (a_-) – yuqoriga (pastga) tezlanish.

Diskka ta'sir etuvchi kuch:

$$F = ma_{\pm} = qE \mp mg = \frac{qV}{d} \mp mg,$$

$mgd \ll qV$ limitida $a_{\pm}$ ni quyidagicha yaqinlashtirish mumkin:

$$a_0 = a_+ = a_- \approx \frac{qV}{md},$$

bu yuqoriga va pastga harakatning simmetrik ekanligini bildiradi. Shunday qilib, (e1) tenglamani $t_0 = t_+ = t_-$, $v_s = v_{0+} = v_{0-}$ va $a_0 = a_+ = a_-$ bilan bitta tenglama orqali tasvirlash mumkin. Bundan tashqari, diskning to'qnashuvdan keyingi tezligi yuqori va pastki plastinkalar uchun bir xil bo'lgani uchun quyidagi munosabatni topamiz:

$$v_s = \eta(v_s + a_0 t_0),$$

bu erdan $\Delta t = 2t_0$ ni olamiz:

$$\Delta t = 2t_0 = 2 \left(\frac{1 - \eta}{\eta} \right) \frac{v_s}{a_0}. \quad (\text{e4})$$

(d6) tenglamadan, $mgd \ll qV$ limitida:

$$K_s = \frac{1}{2}mv_s^2 \approx \left(\frac{\eta^2}{1 - \eta^2} \right) qV. \quad (\text{e6})$$

(e4) va (e6) natijalarini qo'yib:

$$\Delta t = 2 \left(\frac{1 - \eta}{\eta} \right) \sqrt{\frac{2\eta^2}{1 - \eta^2}} \sqrt{\frac{md^2}{qV}} = 2 \sqrt{\frac{1 - \eta}{1 + \eta}} \sqrt{\frac{2md^2}{\chi V^2}}.$$

Shuning uchun $I = \frac{\Delta Q}{\Delta t} = \frac{2q}{\Delta t}$ dan:

$$I = \frac{2q}{\Delta t} = \chi V \sqrt{\frac{1 + \eta}{1 - \eta}} \sqrt{\frac{\chi V^2}{2md^2}} = \sqrt{\frac{1 + \eta}{1 - \eta}} \sqrt{\frac{\chi^3}{2md^2}} V^2.$$

$$\therefore \gamma = \sqrt{\frac{1 + \eta}{1 - \eta}} \sqrt{\frac{\chi^3}{2md^2}}.$$

[Alternativ yechim 1:] (e3) dan boshlab, (e1) ning kvadrat tenglamasini yechamiz.

$$t_{\pm} = \frac{v_{0\pm}}{a_0} \left(\sqrt{1 + \frac{2da_0}{v_{0\pm}^2}} - 1 \right). \quad (\text{e10})$$

Turg'un holatga erishganda, boshlang'ich tezliklar:

$$v_{0\pm} = v_s \quad (\text{e11})$$

$$v_{0-} = \eta \cdot (v_s + a_0 t_+) = \eta v_s \sqrt{1 + \frac{2da_0}{v_s^2}}, \quad (\text{e12})$$

bu yerda v_s ni (e6) natijasidan foydalanib qayta yozish mumkin:

$$v_s^2 \approx \alpha V = \left(\frac{\eta^2}{1 - \eta^2} \right) \frac{2qV}{m} = \left(\frac{\eta^2}{1 - \eta^2} \frac{2a_0 d}{m} \right). \quad (\text{e13})$$

Natijada $v_{0-} \approx \eta v_s \cdot \frac{1}{\eta} = v_s$ va shuning uchun $t_{\pm} = \frac{v_s}{a_0} \left(\frac{1}{\eta} - 1 \right)$ hosil bo'ladi, bu (e4) tenglamaga ekvivalent.

[Alternativ yechim 2:] Tok I ni quyidagicha olish mumkin:

$$I = \frac{2q}{\Delta t} = \frac{2q\bar{v}}{d}, \quad (\text{e14})$$

bu yerda $\bar{v}$ – o'rtacha tezlik. $mgd \ll qV$ limitida yuqoriga va pastga harakatlar bir xil doimiy tezlanish bilan simmetrik bo'lgani uchun:

$$\bar{v} = \frac{1}{2} \left(v_s + \frac{v_s}{\eta} \right). \quad (\text{e15})$$

Shunday qilib,

$$I = \frac{q}{2d} \left(1 + \frac{1}{\eta} \right) v_s. \quad (\text{e16})$$

(e15) dan v_s ifodasini (e16) ga qo'yib, (e8) bilan bir xil ifoda olinadi.

(f) [3 ball]

Disk kinetik energiyasini yo'qotadi va nihoyat harakatdan to'xtaydi, agar disk yuqori plastinkaga yeta olmasa. Boshqacha aytganda, V_c chegaraviy kuchlanishni yuqori plastinkadagi disk tezligi $v_0 = 0$ shartidan aniqlash mumkin.

Yuqori plastinkada $v_0 = 0$ bo'lishi uchun, yuqori plastinkadagi kinetik energiya $\bar{K}_s$ quyidagi munosabatni qanoatlantirishi kerak:

$$\bar{K}_s = K_s + qV_c - mgd = 0, \quad (f1)$$

bu yerda K_s – pastki plastinkadagi to'qnashuvdan keyingi turg'un kinetik energiya. Shunday qilib,

$$\left(\frac{\eta^2}{1-\eta^2}\right)qV_c + \left(\frac{\eta^2}{1+\eta^2}\right)mgd + qV_c - mgd = 0, \quad (f2)$$

yoki ekvivalent ravishda,

$$(1+\eta^2)qV_c - (1-\eta^2)mgd = 0. \quad (f3)$$

$$\therefore qV_c = \frac{1-\eta^2}{1+\eta^2}mgd \quad (f4)$$

$q = \chi V_c$ munosabatidan:

$$\therefore V_c = \sqrt{\frac{1-\eta^2}{1+\eta^2}} \sqrt{\frac{mgd}{\chi}}. \quad (f5)$$

(c4) tenglamadagi V_{th} bilan solishtirganda, (f5) ni quyidagicha yozish mumkin:

$$V_c = z_c V_{th} \quad (f6)$$

bu yerda z_c I ning (V/V_{th}) ga nisbatan grafigida ishlatilishi kerak va

$$z_c = \sqrt{\frac{1-\eta^2}{2(1+\eta^2)}}. \quad (f7)$$

Endi $\Delta t = t_- + t_+$ vaqt oralig'ini aniqlaydigan tenglamalarni tuzamiz:

$$v_{0-}t_- + \frac{1}{2}a_-t_-^2 = d \quad (f8)$$

$$v_{0+}t_+ + \frac{1}{2}a_+t_+^2 = d \quad (f9)$$

bu yerda tezlanishlar:

$$a_+ = \frac{qV_c}{md} - g = \left[\frac{1-\eta^2}{1+\eta^2} - 1\right]g = \left(\frac{-2\eta^2}{1+\eta^2}\right)g \quad (f10)$$

$$a_- = \frac{qV_c}{md} + g = \left[\frac{1 - \eta^2}{1 + \eta^2} + 1 \right] g = \left(\frac{2}{1 + \eta^2} \right) g \quad (\text{f11})$$

$$\frac{a_+}{a_-} = -\eta^2 \quad (\text{f12})$$

$v_{0-} = 0$ bo'lgani uchun $v_{0+} = \eta(a_- t_-)$ va $t_-^2 = 2d/a_-$.

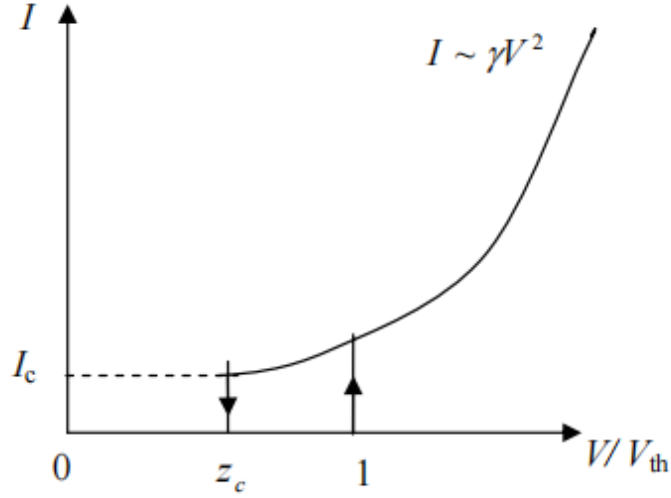
$$t_- = \sqrt{\frac{2d}{a_-}} = \sqrt{(1 + \eta^2) \left(\frac{d}{g} \right)} \quad (\text{f13})$$

$v_{0+}^2 = \eta^2(2da_-) = -2da_+$ dan foydalanib, (f9) kvadrat tenglamasini yechamiz:

$$t_+ = \frac{v_{0+}}{a_+} \left(\sqrt{1 + \frac{2da_+}{v_{0+}^2}} - 1 \right) = \frac{v_{0+}}{a_+} = \sqrt{\frac{2d}{a_+}} = \sqrt{\left(\frac{1 + \eta^2}{\eta^2} \right) \left(\frac{d}{g} \right)} = \frac{t_-}{\eta} \quad (\text{f14})$$

$$\therefore \Delta t = t_- + t_+ = \left(1 + \frac{1}{\eta} \right) \sqrt{(1 + \eta^2) \left(\frac{d}{g} \right)} \quad (\text{f15})$$

$$I_c = \frac{\Delta Q_c}{\Delta t} = \frac{2q}{\Delta t} = \frac{2\chi V_c}{\Delta t} = \frac{2\eta\sqrt{1 - \eta^2}}{(1 + \eta)(1 + \eta^2)} g\sqrt{m\chi} \quad (\text{f16})$$



Tok-kuchlanish xarakteristikasi (kichik grafik)

[Yanada batafsil yechim:] V ning ixtiyoriy qiymati uchun umumiy yechim topish mumkin. (f8) va (f9) kvadrat tenglamalarini yechib:

$$t_{\pm} = \frac{v_{0\pm}}{a_{\pm}} \left[-1 + \sqrt{1 + \frac{2da_{\pm}}{v_{0\pm}^2}} \right]. \quad (\text{f17})$$

Belgilashlarni soddalashtirish uchun bir necha o'zgaruvchi kiritamiz:

$$(i) \quad y = \frac{V}{V_{\text{th}}} \text{ bu yerda } V_{\text{th}} = \sqrt{\frac{2mgd}{\chi}},$$

$$(ii) \quad z_c = \sqrt{\frac{1-\eta^2}{2(1+\eta^2)}}, \text{ (f7) da aniqlangan,}$$

$$(iii) \quad w_0 = 2\eta\sqrt{\frac{gd}{1-\eta^2}} \text{ va } w_1 = 2\sqrt{\frac{d}{(1-\eta^2)g}},$$

y , w va z_c larda:

$$a_+ = \frac{qV}{md} - g = g(2y^2 - 1) \quad (\text{f18})$$

$$a_- = \frac{qV}{md} + g = g(2y^2 + 1) \quad (\text{f19})$$

$$v_{0+} = v_s = w_0\sqrt{y^2 + z_c^2} \quad (\text{f20})$$

$$v_{0-} = \eta(v_s + a_+t_+) = w_0\sqrt{y^2 - z_c^2} \quad (\text{f21})$$

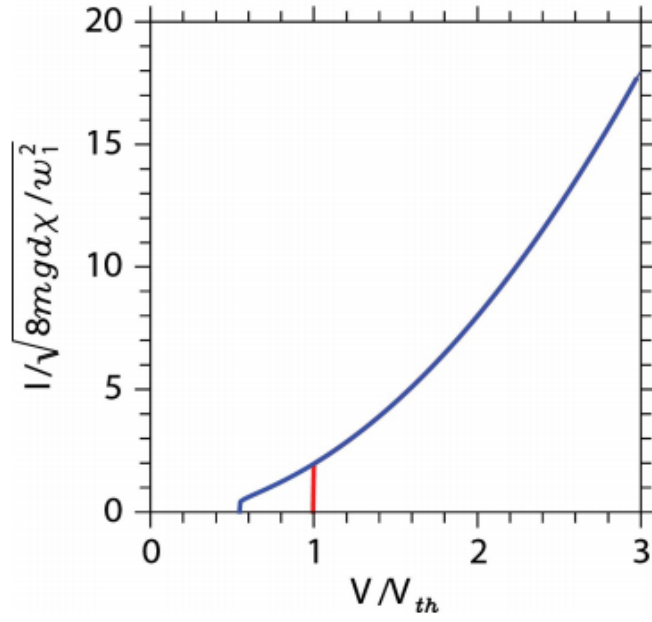
$$t_+ = w_1 \frac{\sqrt{y^2 - z_c^2} - \eta\sqrt{y^2 + z_c^2}}{2y^2 - 1} \quad (\text{f22})$$

$$t_- = w_1 \frac{\sqrt{y^2 + z_c^2} - \eta\sqrt{y^2 - z_c^2}}{2y^2 + 1} \quad (\text{f23})$$

$$I = \frac{\Delta Q}{\Delta t} = \frac{2q}{t_+ + t_-} = (2\chi V_{th}) \frac{y}{\Delta t} = \frac{\sqrt{8mg\chi}}{w_1} F(y) \quad (f24)$$

bu yerda

$$F(y) = y \left\{ \sqrt{y^2 - z_c^2} - \eta \sqrt{y^2 + z_c^2} + \sqrt{y^2 + z_c^2} - \eta \sqrt{y^2 - z_c^2} \right\}^{-1} \quad (f25)$$



I ning V/V_{th} ga bog'liqligi (grafik)

3. Ball taqsimoti

No.	Umumiy ball	Qisman ball	Mazmuni
(a)	1.2	0.3	Gauss qonuni yoki parallel plastinkali kondensator sig'imi formulasi
		0.5	Kondensatorning V kuchlanishdagi to'liq energiyasi / boshqa plastinka maydoni E'
		0.4	Energiya ifodasidan kuchni topish / $F = QE'$
(b)	0.8	0.3	Gauss qonuni / maydon nisbati va (a) natijasidan foydalanish
		0.5	To'g'ri javob
(c)	0.5	0.1	To'g'ri ko'tarilish sharti (kuchlar muvozanati)
		0.2	Maydon nisbati va (a) natijasidan foydalanish
		0.2	To'g'ri javob
(d)	2.3	0.5	Energiya saqlanish va maydon bajargan ish
		0.5	To'qnashuvlarda energiya yo'qotilishi
		0.8	Turg'un holat sharti: energiya balansi tenglamasi (yo'qotish = olish) / rekursiya munosabati
		0.5	To'g'ri javob
(e)	2.2	0.2	Bir aylanishda $\Delta Q = 2q$
		0.5	$qV \gg mgd$ limitida $a_{\pm}$ tezlanishlar; $a_+ = a_-$ simmetriya
		0.3	d, v, a, t uchun kinematik tenglamalar, $t_{\pm}$ yechimlari / simmetriyadan (e4) ni chiqarish
		0.4	Turg'un holatda $v_{0\pm}$ va $t_{\pm}$ ifodalari
		0.4	Yaqinlashuvdagi $t_{\pm}$ yechimlari
		0.4	To'g'ri javob
(f)	3.0	0.5	V_c sharti: $K_{up} = 0$ yoki $v_{s,up} = 0$ / rekursiya munosabati
		0.3	Energiya balansi tenglamasi
		0.3	V_c ning to'g'ri javobi
		0.7	Δt uchun kinematik tenglamalar
		0.3	I_c ning to'g'ri javobi
		0.9	V_{th} va V_c orasidagi farq, asimptotik xulq; grafiklarda $I = \gamma V^2$
Jami	10		

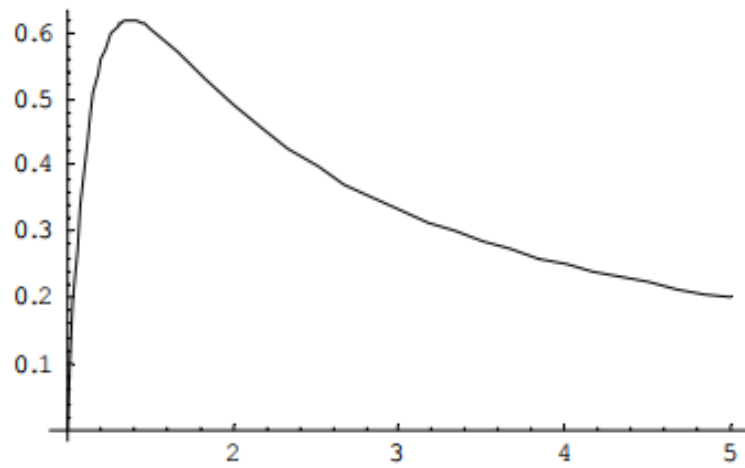
2-nazariy savol: Ko‘tariluvchi shar

1. Javoblar

(a) $F_B = M_A n g \frac{P}{P + \Delta P}$

(b) $\gamma = \frac{\rho_0 z_0 g}{P_0} = 5.5$

(c) $\Delta P = \frac{4\kappa RT}{r_0} \left(\frac{1}{\lambda} - \frac{1}{\lambda^7} \right)$



(d) $a = 0.110$

(e) $z_f = 11 \text{ km}, \quad \lambda_f = 2.1$

2. Yechimlar

A qism

(a) [1.5 ball] Ideal gaz tenglamasidan geliy hajmi $V = nRT/(P + \Delta P)$, shu hajmdagi havoning moli $n' = PV/(RT) = nP/(P + \Delta P)$. Suzuvchi kuch (Arximed kuchi) shu havodagi havoning og'irligiga teng:

$$F_B = M_A n' g = M_A n g \frac{P}{P + \Delta P}. \quad (\text{a3})$$

(b) [2 ball] Bosimning balandlik bilan o'zgarishi: $\frac{dP}{dz} = -\rho g$. Ideal gaz qonunidan $\rho = \frac{M_A P}{RT}$, shuningdek $T = T_0(1 - z/z_0)$. Berilgan $P(z) = P_0(1 - z/z_0)^\gamma$ ni differensiallab va tenglashtirib:

$$\frac{dP}{dz} = -\gamma \frac{P_0}{z_0} \left(1 - \frac{z}{z_0}\right)^{\gamma-1} = -\frac{M_A P}{RT} g = -\frac{M_A P_0}{RT_0} \left(1 - \frac{z}{z_0}\right)^{\gamma-1} g.$$

Demak $\gamma = \frac{M_A g z_0}{RT_0}$. $M_A/RT_0 = \rho_0/P_0$ ekanligidan $\gamma = \frac{\rho_0 g z_0}{P_0}$. Son qiymati:

$$\gamma = \frac{1.16 \times 9.8 \times 4.9 \times 10^4}{1.01 \times 10^5} = 5.52 \approx 5.5.$$

B qism

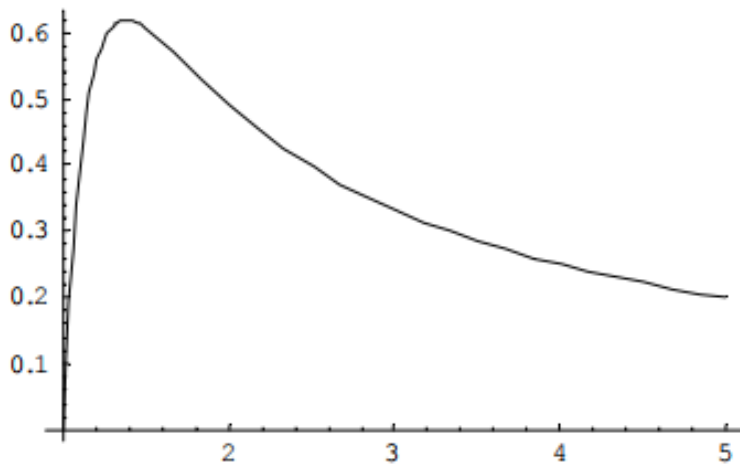
(c) [2 ball] Radiusni r dan $r + dr$ ga oshirishda bajarilgan ish $dW = 4\pi r^2 \Delta P dr$. Elastik energiya $U = 4\pi r_0^2 \kappa RT (2\lambda^2 + \lambda^{-4} - 3)$ dan dU/dr hisoblanadi:

$$\frac{dU}{dr} = 4\pi r_0^2 \kappa RT \left(4\frac{r}{r_0^2} - 4\frac{r_0^6}{r^5} \right) = 4\pi \kappa RT \left(4r - 4\frac{r_0^6}{r^5} \right).$$

$dW = dU$ dan:

$$4\pi r^2 \Delta P dr = 4\pi \kappa RT \left(4r - 4\frac{r_0^6}{r^5} \right) dr \Rightarrow \Delta P = 4\kappa RT \left(\frac{1}{r} - \frac{r_0^6}{r^7} \right) = \frac{4\kappa RT}{r_0} \left(\frac{1}{\lambda} - \frac{1}{\lambda^7} \right).$$

Grafik: $\lambda > 1$ da ΔP avval keskin oshadi, $\lambda = 7^{1/6} \approx 1.38$ da maksimumga erishadi, keyin λ^{-1} kabi kamayadi. $\Delta P/(4\kappa RT/r_0)$ ning λ ga bog'liqligi quyida keltirilgan.



(d) [1.5 ball] Cho'zilmagan holatda $P_0 V_0 = n_0 RT_0$. $\lambda = 1.5$ da hajm $V = \lambda^3 V_0$, ichki bosim $P_{in} = nRT_0/V = \frac{n}{n_0 \lambda^3} P_0$. Shu bilan birga (c) dan $P_{in} = P_0 + \Delta P = P_0 + \frac{4\kappa RT_0}{r_0} \left(\frac{1}{\lambda} - \frac{1}{\lambda^7} \right)$. Tenglab:

$$\frac{n}{n_0 \lambda^3} P_0 = P_0 + \frac{4\kappa RT_0}{r_0} \left(\frac{1}{\lambda} - \frac{1}{\lambda^7} \right).$$

$\kappa_0 = r_0 P_0 / (4RT_0)$ va $a = \kappa / \kappa_0$ ekanligidan $\frac{4\kappa RT_0}{r_0} = a P_0$. Shuning uchun

$$\frac{n}{n_0 \lambda^3} = 1 + a \left(\frac{1}{\lambda} - \frac{1}{\lambda^7} \right) \Rightarrow a = \frac{\frac{n}{n_0 \lambda^3} - 1}{\frac{1}{\lambda} - \frac{1}{\lambda^7}}.$$

$n/n_0 = 3.6$, $\lambda = 1.5$ ni qo'yib: $a = 0.110$.

$$a = \frac{n/(n_0\lambda^3) - 1}{\lambda^{-1} - \lambda^{-7}}. \quad (d5)$$

$n/n_0 = 3.6$ va $\lambda = 1.5$ ni qo'yib, $a = 0.110$.

C qism

(e) [3 ball] (a) dagi suzuvchi kuch umumiy massa $M_T = 1.12$ kg ni muvozanatlashi kerak. Shunday qilib, (a3) tenglamadan og'irlik muvozanati sharti:

$$\frac{P}{P + \Delta P} = \frac{M_T}{M_A n}. \quad (e1)$$

Boshqa tomondan, $V = \frac{4}{3}\pi r^3 = \lambda^3 V_0$ hajmdagi geliy gaziga ideal gaz qonunini qo'llab, ixtiyoriy P va T uchun:

$$(P + \Delta P)\lambda^3 = \frac{nRT}{V_0} = P_0 \frac{T}{T_0} \frac{n}{n_0}. \quad (e2)$$

(c3), (e1) va (e2) tenglamalar P , ΔP va λ ni T va boshqa parametrlar orqali aniqlaydi. (e2) ni (e1) da ishlatib, og'irlik muvozanati uchun alternativ shart:

$$\frac{P}{P_0} \frac{T_0}{T} \lambda^3 = \frac{M_T}{M_A n_0}. \quad (e3)$$

Endi (c3) dan ΔP ni (e2) ga qo'yib:

$$P\lambda^3 + \frac{4\kappa RT}{r_0} \lambda^2 (1 - \lambda^{-6}) = P_0 \frac{T}{T_0} \frac{n}{n_0}$$

yoki qayta tartibga solib,

$$\frac{P}{P_0} \frac{T_0}{T} \lambda^3 = \frac{n}{n_0} - a\lambda^2 (1 - \lambda^{-6}), \quad (e4)$$

bu yerda a ta'rifi yana ishlatilgan. (e3) va (e4) ning o'ng tomonlarini tenglashtirib, λ uchun tenglama hosil qilamiz:

$$\lambda^2 (1 - \lambda^{-6}) = \frac{1}{an_0} \left(n - \frac{M_T}{M_A} \right) = 4.54. \quad (e5)$$

λ ning yechimi:

$$\lambda^2 \approx 4.54 / (1 - 4.54^{-3}) \approx 4.54 : \lambda_f \approx 2.13. \quad (e6)$$

Balandlikni topish uchun (e3) tenglamaning chap tomonidagi $(P/P_0)/(T/T_0)$ ni (b) da berilgan balandlik funksiyasi bilan almashtiramiz:

$$\frac{P}{P_0} \frac{T_0}{T} \lambda^3 = (1 - z_f/z_0)^{\gamma-1} \lambda_f^3 = \frac{M_T}{M_A n_0} = 3.10. \quad (\text{e7})$$

$\lambda_f = 2.13$ va $\gamma - 1 = 4.5$ bilan z_f uchun (e7) tenglamani yechamiz:

$$z_f = 49 \times (1 - (3.10/2.13^3)^{1/4.5}) = 10.9 \text{ (km)}.$$

Talab qilingan javoblar: $\lambda_f = 2.1$, $z_f = 11 \text{ km}$.

3. Ball taqsimoti

No.	Umumiy ball	Qisman ball	Mazmuni
(a)	1.5	0.5 0.5 0.5	Arximed qonuni Ideal gaz qonunini to'g'ri qo'llash To'g'ri javob (He og'irligini ayirish uchun 0.3)
(b)	2.0	0.8 0.5 0.5 0.2	Bosim farqining havo zichligiga aloqasi Zichlikni bosimga aylantirish uchun ideal gaz qonunini qo'llash γ uchun to'g'ri formula Javobdagi to'g'ri son
(c)	2.0	0.7 0.3 0.5 0.5	Mexanik ishning elastik energiya o'zgarishiga aloqasi Bosimning kuchga aloqasi Formuladagi to'g'ri javob Egri chiziqning to'g'ri eskizi
(d)	1.5	0.3 0.4 0.5 0.3	Ichki bosim uchun ideal gaz qonunini qo'llash Berilgan sharoitlarda ichki bosimni a orqali ifodalash a uchun formula yoki to'g'ri ifoda To'g'ri javob
(e)	3.0	0.3 0.3 0.5 0.7 0.5 0.7	Noma'lumlarni aniqlash uchun kuchlar muvozanatidan foydalanish Gazga qo'llanilgan ideal gaz qonuni mustaqil shart sifatida λ_f ni sonli aniqlash sharti λ_f uchun to'g'ri javob z_f va λ_f o'rtasidagi bog'liqlik z_f uchun to'g'ri javob
Jami	10		

3-nazariy savol: Skanerli zond mikroskopi (SPM)

1. Javoblar

(a) $A = \frac{F_0}{\sqrt{m^2(\omega_0^2 - \omega^2)^2 + b^2\omega^2}}$ va $\tan \phi = \frac{b\omega}{m(\omega_0^2 - \omega^2)}$. $\omega = \omega_0$ da $A = \frac{F_0}{b\omega_0}$ va $\phi = \frac{\pi}{2}$.

(b) Nolga teng bo'lmagan doimiy tok komponentasi faqat $\omega = \omega_i$ da mavjud. Bu holda doimiy tok signalining amplitudasi $\frac{1}{2}V_{i0}V_{R0} \cos \phi_i$ ga teng.

(c) $\frac{c_1 c_2}{2} \frac{V_{R0}^2}{b\omega_0}$ (rezonans chastotasi ω_0 da).

(d) $\Delta m = 1.7 \times 10^{-18}$ kg.

(e) $\omega'_0 = \omega_0 \left(1 - \frac{c_3}{m\omega_0^2}\right)^{1/2}$.

(f) $d_0 = \left(k_e \frac{qQ}{m\omega_0 \Delta\omega_0}\right)^{1/3}$ va $d_0 = 41$ nm.

2. Yechimlar

(a) [1.5 ball] $z(t) = A \sin(\omega t - \phi)$ ni $m \frac{d^2 z}{dt^2} + b \frac{dz}{dt} + m\omega_0^2 z = F_0 \sin \omega t$ tenglamasiga qo'yib:

$$-m\omega^2 \sin(\omega t - \phi) + b\omega \cos(\omega t - \phi) + m\omega_0^2 \sin(\omega t - \phi) = \frac{F_0}{A} \sin \omega t. \quad (\text{a1})$$

$\sin \omega t$ va $\cos \omega t$ ga proporsional hadlarni yig'ib:

$$\left\{ m(\omega_0^2 - \omega^2) \cos \phi + b\omega \sin \phi - \frac{F_0}{A} \right\} \sin \omega t + \left\{ -m(\omega_0^2 - \omega^2) \sin \phi + b\omega \cos \phi \right\} \cos \omega t = 0 \quad (\text{a2})$$

Har bir jingalak qavsni nolga tenglashtirib:

$$\tan \phi = \frac{b\omega}{m(\omega_0^2 - \omega^2)}, \quad (\text{a3})$$

$$A = \frac{F_0}{\sqrt{m^2(\omega_0^2 - \omega^2)^2 + b^2\omega^2}}. \quad (\text{a4})$$

$\omega = \omega_0$ da:

$$A = \frac{F_0}{b\omega_0} \quad \text{va} \quad \phi = \frac{\pi}{2}. \quad (\text{a5})$$

(b) [1 ball] Ko'paytirilgan signal:

$$V_{i0} \sin(\omega_i t - \phi_i) V_{R0} \sin(\omega t) = \frac{1}{2} V_{i0} V_{R0} [\cos\{(\omega_i - \omega)t - \phi_i\} - \cos\{(\omega_i + \omega)t - \phi_i\}] \quad (\text{b1})$$

Nolga teng bo'lmagan doimiy tok komponentasi faqat $\omega = \omega_i$ da mavjud. Bu holda doimiy tok signalining amplitudasi $\frac{1}{2} V_{i0} V_{R0} \cos \phi_i$ ga teng. (b2)

(c) [1.5 ball] Lock-in kuchaytirgichi bir xil chastotadagi o'zgaruvchan tok signalini mos yozuv signali bilan o'lgahanligi sababli, piezoelektrik trubaning tebranish chastotasi, konsol chastotasi va fotodiod detektorining chastotasi bir xil bo'lishi kerak. Rezonansdagi kirish signalining kattaligi:

$$V_{io} = c_2 \frac{F_0}{b\omega_0} = \frac{c_1 c_2 V_{R0}}{b\omega_0}. \quad (c1)$$

Rezonansda kirish signalining fazasi $-\frac{\pi}{2} + \frac{\pi}{2} = 0$ ga teng, shuning uchun $\phi_i = 0$ va lock-in kuchaytirgich signali:

$$\frac{1}{2} V_{io} V_{R0} \cos 0 = \frac{c_1 c_2 V_{R0}^2}{2b\omega_0}. \quad (c2)$$

(d) [2 ball] Dastlabki rezonans chastotasi $\omega_0 = \sqrt{\frac{k}{m}}$ quyidagicha o'zgaradi:

$$\sqrt{\frac{k}{m + \Delta m}} = \sqrt{\frac{k}{m} \left(1 + \frac{\Delta m}{m}\right)^{-1/2}} \approx \sqrt{\frac{k}{m} \left(1 - \frac{1}{2} \frac{\Delta m}{m}\right)} = \omega_0 \left(1 - \frac{1}{2} \frac{\Delta m}{m}\right). \quad (d1)$$

Shunday qilib,

$$\Delta\omega_0 = -\frac{1}{2}\omega_0 \frac{\Delta m}{m}. \quad (d2)$$

Rezonans yaqinida, (a3) tenglamaga $\phi \rightarrow \frac{\pi}{2} + \Delta\phi$ va $\omega_0 \rightarrow \omega_0 + \Delta\omega_0$ ni qo'yib, ω_0 ning kichik o'zgarishi tufayli faza o'zgarishi:

$$\tan\left(\frac{\pi}{2} + \Delta\phi\right) = -\frac{1}{\tan \Delta\phi} = -\frac{b}{2m\Delta\omega_0}. \quad (d3)$$

Shuning uchun,

$$\Delta\phi \approx \tan \Delta\phi = -\frac{2m\Delta\omega_0}{b}. \quad (d4)$$

(d2) va (d4) dan:

$$\Delta m = \frac{b}{\omega_0} \Delta\phi = \frac{10^3 \cdot 10^{-12}}{10^6} \frac{\pi}{1800} = \frac{\pi}{1.8} \cdot 10^{-18} = 1.7 \cdot 10^{-18} \text{ kg}. \quad (d5)$$

(e) [1.5 ball] O‘zaro ta‘sir mavjud bo‘lganda, yangi muvozanat holati h_0 atrofidagi harakat tenglamasi:

$$m \frac{d^2 z}{dt^2} + b \frac{dz}{dt} + m \omega_0^2 z - c_3 z = F_0 \sin \omega t \quad (\text{e1})$$

bu yerda $f(h) \approx f(h_0) + c_3 z$ va $z = h - h_0$ yangi muvozanat holatidan siljish. $f(h_0)$ doimiy hadi yangi muvozanat holatida qisqaradi. Shunday qilib, dastlabki rezonans chastotasi $\omega_0 = \sqrt{k/m}$ quyidagicha siljiydi:

$$\omega'_0 = \sqrt{\frac{k - c_3}{m}} = \sqrt{\frac{m \omega_0^2 - c_3}{m}} = \omega_0 \sqrt{1 - \frac{c_3}{m \omega_0^2}} \quad (\text{e3})$$

Demak, rezonans chastotasining siljishi:

$$\Delta \omega_0 = \omega_0 \left[\sqrt{1 - \frac{c_3}{m \omega_0^2}} - 1 \right] \quad (\text{e4})$$

(f) [2.5 ball] Maksimal siljish konsol zaryad ustida turganda sodir bo‘ladi, bunda o‘zaro ta‘sir kuchi:

$$f(h) = k_e \frac{qQ}{h^2} \quad (\text{f1})$$

Bu yerdan,

$$c_3 = \left. \frac{df}{dh} \right|_{h=d_0} = -2k_e \frac{qQ}{d_0^3} \quad (\text{f2})$$

$\Delta \omega_0 \ll \omega_0$ bo‘lgani uchun (e4) ni quyidagicha yaqinlashtirish mumkin:

$$\Delta \omega_0 \approx -\frac{c_3}{2m \omega_0} \quad (\text{f3})$$

(f2) va (f3) dan:

$$\Delta \omega_0 = -\frac{1}{2m \omega_0} \left(-2k_e \frac{qQ}{d_0^3} \right) = k_e \frac{qQ}{m \omega_0 d_0^3} \quad (\text{f4})$$

Bu yerda $q = e = -1.6 \times 10^{-19}$ C va $Q = 6e = -9.6 \times 10^{-19}$ C.

$$d_0 = \left(k_e \frac{qQ}{m\omega_0 \Delta\omega_0} \right)^{1/3} = 4.1 \times 10^{-8} \text{ m} = 41 \text{ nm}. \quad (\text{f5})$$

Shunday qilib, tuzoqqa tushgan elektron konsoldan 41 nm masofada joylashgan.

3. Ball taqsimoti

No.	Umumiy ball	Qisman ball	Mazmuni
(a)	1.5	0.7	A va ϕ uchun tenglamalar (almashtirish va o'zgartirish)
		0.4	A va ϕ uchun to'g'ri javoblar
		0.4	ω_0 dagi A va ϕ
(b)	1.0	0.4	Ko'paytirilgan signal tenglamasi
		0.3	Nolga teng bo'lmagan doimiy tok chiqishi sharti
		0.3	Doimiy tok chiqishi uchun to'g'ri javob
(c)	1.5	0.6	V_i va V_R o'rtasidagi bog'liqlik
		0.4	Maksimal doimiy tok chiqishi sharti
		0.5	Doimiy tok chiqishining kattaligi uchun to'g'ri javob
(d)	2.0	0.5	Δm va $\Delta\omega_0$ o'rtasidagi bog'liqlik
		1.0	$\Delta\omega_0$ (yoki Δm) va $\Delta\phi$ o'rtasidagi bog'liqliklar
		0.5	To'g'ri javob (noto'g'ri ishora uchun 0.2)
(e)	1.5	1.0	$f(h)$ bilan tenglamani o'zgartirish va tegishli yaqinlashuvdan foydalanish
		0.5	To'g'ri javob
(f)	2.5	0.5	Kulon kuchining to'g'ri formulasi-dan foydalanish
		0.3	c_3 ni hisoblash
		0.6	(e) natijasidan $\Delta\omega_0$ yoki $\omega_0^2 - \omega'^2_0$ uchun foydalanish
		0.6	d_0 uchun ifoda
		0.5	To'g'ri javob
Jami	10		

Eksperimental qism: Mexanik “Qora quti” (MBB)

A-QISM: Sharning massasi va joylashuvi ko‘paytmasi ($m \times l$) (4.0 ball)

1. $m \times l$ ni olish imkonini beruvchi usulni tenglamalar yordamida taklif qiling va asoslang. (2.0 ball)

$$m \times l = (M + m) \times l_{\text{cm}}$$

(Tushuntirish) Butun tizimning massa markazi joylashuvi topilgach, mexanik “Qora quti”ga richag qoidasi (lever rule) qo‘llaniladi.

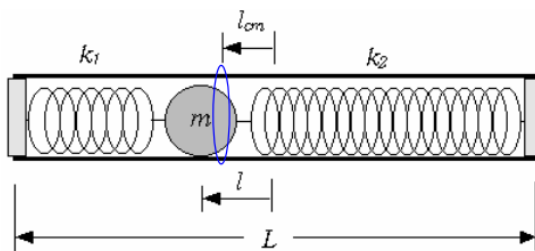


Fig. A-1 Experimental setup

Fig. A-1 Eksperimental qurilma

2. $m \times l$ ning qiymatini eksperimental ravishda aniqlang. (2.0 ball)

$$m \times l = 2.96 \times 10^{-3} \text{ kg}\cdot\text{m}$$

(Tushuntirish) O‘lchangan kattaliklar:

$$M + m = (1.411 \pm 0.0005) \times 10^{-1} \text{ kg}$$

va

$$l_{\text{cm}} = (2.1 \pm 0.06) \times 10^{-2} \text{ m} \quad \text{yoki} \quad 21 \pm 0.6 \text{ mm}.$$

Shuning uchun

$$m \times l = (M + m) \times l_{\text{cm}} = (1.411 \pm 0.0005) \times 10^{-1} \text{ kg} \times (2.1 \pm 0.06) \times 10^{-2} \text{ m} = (2.96 \pm 0.08) \times 10^{-3} \text{ kg}\cdot\text{m}.$$

B-QISM: Sharning massasi m (10.0 ball)

1. Og'irlikning turli tushish masofalari h uchun v ni o'lchang. Ma'lumotlarni m qiymatini topishga qulay shaklda grafik qog'oziga chizing. Grafikda sekin aylanish mintaqasini va tez aylanish mintaqasini aniqlang. (4.0 ball)
2. O'lchovlaringizdan sekin aylanish mintaqasida $h = Cv^2$ va tez aylanish mintaqasida $h = Av^2 + B$ ekanligini ko'rsating. (1.0 ball)

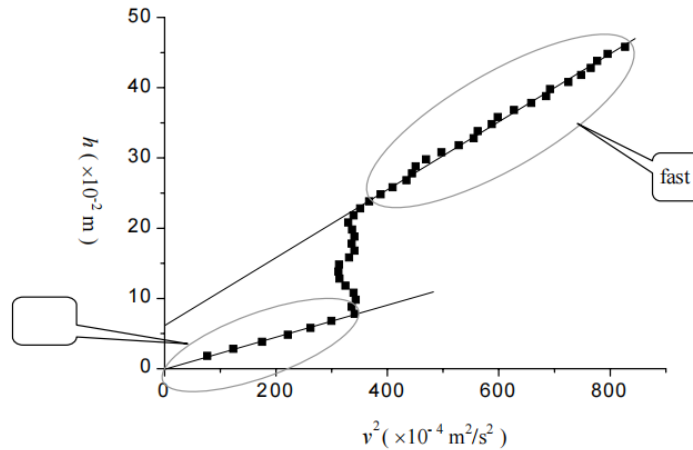


Fig. B-1 Experimental data

Fig. B-1 Eksperimental ma'lumotlar

(Tushuntirish) O'lchangan ma'lumotlar jadvali berilgan.

(O'lchov jadvalining davomi)

h_1 (10^{-2} m)	Δt (ms)	h (10^{-2} m)	v (10^{-2} m/s)	v^2 (10^{-4} m ² /s ²)
33.5±0.1	141.4±0.05	9.8±0.1	18.53±0.04	343±1
34.5±0.1	142.2±0.05	10.8±0.1	18.42±0.04	339±1
35.5±0.1	145.4±0.05	11.8±0.1	18.02±0.04	325±1
...	...	...	...	...
69.5±0.1	91.1±0.05	45.8±0.1	28.76±0.06	827±2

bu yerda h_1 – og‘irlik tusha boshlashdan oldingi yuqori holat ko‘rsatkichi, $h = h_1 - h_2 + d/2$ – og‘irlikning tushish masofasi, $h_2 = (25 \pm 0.05) \times 10^{-2}$ m – fotodarvoza to‘silishi boshlangan paytdagi og‘irlikning yuqori holati, $d = (2.62 \pm 0.005) \times 10^{-2}$ m – og‘irlik uzunligi, $v = d/\Delta t$.

3. C koeffitsientini MBB parametrlari bilan bog‘lang. (1.0 ball)

$$h = Cv^2, \text{ bu yerda } C = \frac{m_0 + I/R^2 + m(l^2 + \frac{2}{5}r^2)/R^2}{2m_0g}$$

(Tushuntirish) Shar statik muvozanatda ($x = l$). Og‘irlik tezligi v bo‘lganda, butun tizimning kinetik energiyasining ortishi:

$$\Delta K = \frac{1}{2}m_0v^2 + \frac{1}{2}I\omega^2 + \frac{1}{2}m(l^2 + \frac{2}{5}r^2)\omega^2 = \frac{1}{2} \left\{ m_0 + \frac{I}{R^2} + \frac{m(l^2 + \frac{2}{5}r^2)}{R^2} \right\} v^2,$$

bu yerda $\omega = v/R$ – MBB ning burchak tezligi, I – shardan tashqari butun tizimning effektiv inersiya momenti. Og‘irlikning gravitatsion potensial energiyasining kamayishi $\Delta U = -m_0gh$ bo‘lgani uchun energiya saqlanish ($\Delta K + \Delta U = 0$) dan yuqoridagi ifoda kelib chiqadi.

4. A va B koeffitsientlarini MBB parametrlari bilan bog‘lang. (1.0 ball)

$$h = Av^2 + B, \text{ bu yerda } A = \frac{m_0 + I/R^2 + m\{(L/2 - \delta - r)^2 + \frac{2}{5}r^2\}/R^2}{2m_0g},$$

$$B = \frac{-k_1(L/2 - l - \delta - r)^2 + k_2\{(L - 2\delta - 2r)^2 - (L/2 + l - \delta - r)^2\}}{2m_0g}$$

(Tushuntirish) Shar trubka qopqog‘ida turadi ($x = L/2 - \delta - r$). Prujinalarning elastik potensial energiyasining ortishi ΔU_e hisobga olinadi.

5. O‘lchovlaringiz va A-QISM natijalaridan foydalanib, m ning qiymatini aniqlang. (3.0 ball)

$$m = 6.2 \times 10^{-2} \text{ kg}$$

(Tushuntirish) B-QISM 3 va 4 natijalaridan:

$$A - C = \frac{m}{2gm_0R^2} \left\{ \left(\frac{L}{2} - \delta - r \right)^2 - l^2 \right\}.$$

O'lchangan qiymatlar:

$$L = (40.0 \pm 0.05) \times 10^{-2} \text{ m}, \quad m_0 = (100.4 \pm 0.05) \times 10^{-3} \text{ kg}, \quad 2R = (3.91 \pm 0.005) \times 10^{-2} \text{ m}.$$

Shuning uchun $(L/2 - \delta - r)^2 = (338.6 \pm 0.8) \times 10^{-4} \text{ m}^2$ va $2gm_0R^2 = (752 \pm 2) \times 10^{-6} \text{ kg} \cdot \text{m}^3/\text{s}^2$.

Grafikdagi ikki to'g'ri chiziqning qiyaliklari:

$$A = 5.0 \pm 0.1 \text{ s}^2/\text{m}, \quad C = 2.4 \pm 0.1 \text{ s}^2/\text{m}, \quad A - C = 2.6 \pm 0.1 \text{ s}^2/\text{m}.$$

A-QISM dan $m \times l = (M + m) \times l_{\text{cm}} = 2.96 \times 10^{-3} \text{ kg} \cdot \text{m}$ ni qo'llab, m uchun kvadrat tenglama hosil qilinadi va uning fizik ma'noli musbat ildizi $m = (6.2 \pm 0.2) \times 10^{-2} \text{ kg}$ chiqadi.

The slopes of the two straight lines in the graph (Fig. B-1) of PART-B 1 are

$$A = 5.0 \pm 0.1 \text{ s}^2/\text{m} \quad \text{and} \quad C = 2.4 \pm 0.1 \text{ s}^2/\text{m},$$

respectively, and

$$A - C = 2.6 \pm 0.1 \text{ s}^2/\text{m}.$$

Since we already obtained $m \times l = (M + m) \times l_{\text{cm}} = 2.96 \times 10^{-3} \text{ kg} \cdot \text{m}$ from PART-A, the equation

$$(338.6 \pm 0.8)m^2 - (752 \pm 2) \times 10^3 \times (0.026 \pm 0.001)m - (296 \pm 8)^2 = 0$$

or

$$(338.6 \pm 0.8)m^2 - (19600 \pm 800)m - (88000 \pm 3000) = 0$$

is resulted, where m is expressed in the unit of g. The roots of this equation are

$$m = \frac{(9800 \pm 400) \pm \sqrt{(9800 \pm 400)^2 + (338.6 \pm 0.8) \times (88000 \pm 3000)}}{(338.6 \pm 0.8)}.$$

The physically meaningful positive root is

$$m = \frac{(9800 \pm 400) + \sqrt{(126000000 \pm 6000000)}}{(338.6 \pm 0.8)} = (62 \pm 2) \text{ g} = (6.2 \pm 0.2) \times 10^{-2} \text{ kg}.$$

C-QISM: Prujina doimiylari k_1 va k_2 (6.0 ball)

1. 3-rasm (1) va (2) da ko'rsatilgan kichik tebranish davrlari T_1 va T_2 ni o'lchang va ularning qiymatlarini yozing. (1.0 ball)

$$T_1 = 1.1090 \text{ s} \quad \text{va} \quad T_2 = 1.0193 \text{ s}$$

(Tushuntirish)

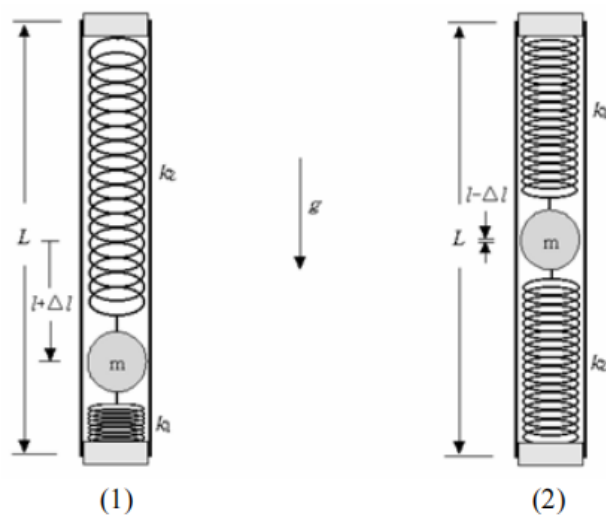


Fig. C-1 Small oscillation experimental set up

Fig. C-1 Kichik tebranish eksperimental qurilmasi

O'lchangan davrlar:

	T_1 (s)	T_2 (s)
1	1.1085 ± 0.00005	1.0194 ± 0.00005
2	1.1092 ± 0.00005	1.0194 ± 0.00005
3	1.1089 ± 0.00005	1.0193 ± 0.00005
4	1.1085 ± 0.00005	1.0191 ± 0.00005
5	1.1094 ± 0.00005	1.0192 ± 0.00005
6	1.1090 ± 0.00005	1.0194 ± 0.00005
7	1.1088 ± 0.00005	1.0194 ± 0.00005
8	1.1090 ± 0.00005	1.0191 ± 0.00005
9	1.1092 ± 0.00005	1.0192 ± 0.00005
10	1.1094 ± 0.00005	1.0193 ± 0.00005

O'rtacha qiymatlar:

$$T_1 = 1.1090 \pm 0.0003 \text{ s}, \quad T_2 = 1.0193 \pm 0.0001 \text{ s}.$$

2. Nima uchun konfiguratsiyalarning kichik tebranish burchak chastotalari ω_1 va ω_2 har xil ekanligini tenglamalar yordamida tushuntiring. (1.0 ball)

$$\omega_1 = \sqrt{\frac{Mg\frac{L}{2} + mg\left(\frac{L}{2} + l + \Delta l\right)}{I_o + m\left(\frac{L}{2} + l + \Delta l\right)^2 + \frac{2}{5}r^2}}$$

$$\omega_2 = \sqrt{\frac{Mg\frac{L}{2} + mg\left(\frac{L}{2} - l + \Delta l\right)}{I_o + m\left(\frac{L}{2} - l + \Delta l\right)^2 + \frac{2}{5}r^2}}$$

(Tushuntirish) MBB ning trubka yuqori qismidagi aylanish o'qiga nisbatan inersiya momenti:

$$I_1 = I_o + m\left(\frac{L}{2} + l + \Delta l\right)^2 + \frac{2}{5}r^2 \quad \text{yoki} \quad I_2 = I_o + m\left(\frac{L}{2} - l + \Delta l\right)^2 + \frac{2}{5}r^2$$

MBB vertikalidan kichik θ burchakka og'ganda, gravitatsiya momenti:

$$\tau_1 = Mg\left(\frac{L}{2}\right)\sin\theta + mg\left(\frac{L}{2} + l + \Delta l\right)\sin\theta \approx \left\{Mg\frac{L}{2} + mg\left(\frac{L}{2} + l + \Delta l\right)\right\}\theta$$

yoki τ_2 shunga o'xshash. Shuning uchun tebranishlarning burchak chastotalari yuqoridagi kabi ifodalanadi.

3. I_0 ni oldingi natijalardan yo'qotib, Δl ni hisoblang. (1.0 ball)

$$\Delta l = (7.2 \pm 0.9) \times 10^{-2} \text{ m}$$

(Tushuntirish) Ikki burchak chastotasi ifodalarini qayta yozib, noma'lum I_0 ni yo'qotish orqali Δl uchun tenglama hosil qilinadi. O'lchangan va berilgan qiymatlardan foydalanib, Δl hisoblanadi.

4. C-QISM 1–3 va B-QISM natijalarini birlashtirib, ikki prujinali tizimning effektiv umumiy prujina doimiysi k ning qiymatini toping va yozing. (2.0 ball)

$$k = 9 \text{ N/m}$$

(Tushuntirish) $k = \frac{mg}{\Delta l} = \frac{(62 \pm 2) \times 980}{7.2 \pm 0.9} = 9000 \pm 1000 \text{ dyne/cm} = 9 \pm 1 \text{ N/m}$.

5. k_1 va k_2 ning mos qiymatlarini oling. (1.0 ball)

$$k_1 = 5.7 \text{ N/m}, \quad k_2 = 3 \text{ N/m}$$

(Tushuntirish) MBB gorizonta tekislikda muvozanatda bo'lganda, shar uchun kuchlar muvozanati sharti:

$$\frac{L/2 - l - \delta - r}{L/2 + l - \delta - r} = \frac{N_1}{N_2} = \frac{k_2}{k_1}.$$

$k = k_1 + k_2$ ekanligidan,

$$k_1 = \frac{k}{\frac{L/2 - l - \delta - r}{L/2 + l - \delta - r} + 1} = \frac{L/2 + l - \delta - r}{L - 2\delta - 2r} k$$

va

$$k_2 = k - k_1 = \frac{L/2 - l - \delta - r}{L - 2\delta - 2r} k.$$

O'lchangan qiymatlardan:

$$\frac{L/2 + l - \delta - r}{L - 2\delta - 2r} = \frac{(20.0 \pm 0.03) + \left(\frac{296 \pm 8}{62 \pm 2}\right) - 0.5 - 1.1}{(40.0 \pm 0.05) - 1.0 - 2.2} = 0.63 \pm 0.005.$$

Shuning uchun

$$k_1 = (0.63 \pm 0.005) \times (9000 \pm 1000) = 5700 \pm 600 \text{ dyne/cm} = 5.7 \pm 0.6 \text{ N/m},$$

$$k_2 = (9000 \pm 1000) - (5700 \pm 600) = 3000 \pm 1000 \text{ dyne/cm} = 3 \pm 1 \text{ N/m}.$$

(Tushuntirish) MBB gorizonta tekislikda muvozanatda bo'lganda, shar uchun kuchlar muvozanati sharti:

$$\frac{L/2 - l - \delta - r}{L/2 + l - \delta - r} = \frac{N_1}{N_2} = \frac{k_2}{k_1}.$$

$k = k_1 + k_2$ ekanligidan,

$$k_1 = \frac{k}{\frac{L/2 - l - \delta - r}{L/2 + l - \delta - r} + 1} = \frac{L/2 + l - \delta - r}{L - 2\delta - 2r} k$$

va

$$k_2 = k - k_1 = \frac{L/2 - l - \delta - r}{L - 2\delta - 2r} k.$$

O'lchangan qiymatlardan:

$$\frac{L/2 + l - \delta - r}{L - 2\delta - 2r} = \frac{(20.0 \pm 0.03) + \left(\frac{296 \pm 8}{62 \pm 2}\right) - 0.5 - 1.1}{(40.0 \pm 0.05) - 1.0 - 2.2} = 0.63 \pm 0.005.$$

Shuning uchun

$$k_1 = (0.63 \pm 0.005) \times (9000 \pm 1000) = 5700 \pm 600 \text{ dyne/cm} = 5.7 \pm 0.6 \text{ N/m},$$

$$k_2 = (9000 \pm 1000) - (5700 \pm 600) = 3000 \pm 1000 \text{ dyne/cm} = 3 \pm 1 \text{ N/m}.$$